

Illustrative mouse head gallery board

Nominal fabrication report

Illustrative planning output. Not a machine program, safety assessment, joint-strength certificate or shop-ready instruction.

Validation: nominal_valid

Finished X / Y / Z: 290 mm / 290 mm / 30 mm

N2 exact deterministic compiler

Nominal micrometre dimensions

Validation coverage

Passed: strict_schema, topological_operations, exact_geometry, source_provenance_partition, terminal_disposition, proper_rotations, glue_compatibility, construction_template, final_dimensions, end_grain_exposure, increment, declared_workpiece_capacity, stage_specific_kerfs, minimum_handling_dimensions, terminal_slicing_reserve, rough_stock_preparation_allowances, operation_derived_face_preparation, fixed_stock_cross_sections, source_separation, uniform_grid, physical_panel_assignments, visual_recipe_cells, surfacing_process_capabilities

Not evaluated: interval_uncertainty_propagation, physical_machine_safety, joint_strength, finite_inventory, optimization

Material accounting

Species	Material category	Volume (mm3)
maple	stock	62,719,272.000
maple	finished	2,035,500.000
maple	kerf	2,500,392.000
maple	milling	8,599,650.000
maple	trim	2,967,972.000
maple	other offcuts	46,615,758.000
walnut	stock	23,161,482.000
walnut	finished	487,500.000
walnut	kerf	853,242.000
walnut	milling	3,674,690.000
walnut	trim	1,050,282.000
walnut	other offcuts	17,095,768.000

Recorded operation sequence

1. maple-1-separate — cut

Inputs: maple-1

Outputs: maple-1-window, maple-1-reserve

maple-1-window: 108 × 395 × 33 mm (X × Y × Z)

maple-1-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

2. maple-1-end-min — cut

Inputs: maple-1-window

Outputs: maple-1-end1, maple-1-end-min-off

maple-1-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-1-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

`{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}`

3. maple-1-end-max — cut

Inputs: maple-1-end1

Outputs: maple-1-blank, maple-1-end-max-off

maple-1-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-1-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

`{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}`

4. maple-1-zmin — surface

Inputs: maple-1-blank

Outputs: maple-1-z1, maple-1-zmin-loss

maple-1-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-1-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

`{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}`

5. maple-1-zmax — surface

Inputs: maple-1-z1

Outputs: maple-1-z2, maple-1-zmax-loss

maple-1-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-1-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

`{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}`

6. maple-1-xmin — surface

Inputs: maple-1-z2

Outputs: maple-1-x1, maple-1-xmin-loss

maple-1-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-1-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

`{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}`

7. maple-1-xmax — surface

Inputs: maple-1-x1

Outputs: maple-1-x2, maple-1-xmax-loss

maple-1-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-1-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

`{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}`

8. maple-1-rip — cut

Inputs: maple-1-x2

Outputs: maple-1-pre, maple-1-rip-off

maple-1-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-1-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

9. maple-1-joint — surface

Inputs: maple-1-pre

Outputs: panel-1-strip-0, maple-1-joint-loss

panel-1-strip-0: 25 × 370 × 27 mm (X × Y × Z)

maple-1-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

10. maple-2-separate — cut

Inputs: maple-2

Outputs: maple-2-window, maple-2-reserve

maple-2-window: 108 × 395 × 33 mm (X × Y × Z)

maple-2-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

11. maple-2-end-min — cut

Inputs: maple-2-window

Outputs: maple-2-end1, maple-2-end-min-off

maple-2-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-2-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

12. maple-2-end-max — cut

Inputs: maple-2-end1

Outputs: maple-2-blank, maple-2-end-max-off

maple-2-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-2-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

13. maple-2-zmin — surface

Inputs: maple-2-blank

Outputs: maple-2-z1, maple-2-zmin-loss

maple-2-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-2-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

14. maple-2-zmax — surface

Inputs: maple-2-z1

Outputs: maple-2-z2, maple-2-zmax-loss

maple-2-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-2-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

15. maple-2-xmin — surface

Inputs: maple-2-z2

Outputs: maple-2-x1, maple-2-xmin-loss

maple-2-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-2-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

16. maple-2-xmax — surface

Inputs: maple-2-x1

Outputs: maple-2-x2, maple-2-xmax-loss

maple-2-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-2-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

17. maple-2-rip — cut

Inputs: maple-2-x2

Outputs: maple-2-pre, maple-2-rip-off

maple-2-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-2-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

18. maple-2-joint — surface

Inputs: maple-2-pre

Outputs: panel-1-strip-1, maple-2-joint-loss

panel-1-strip-1: 25 × 370 × 27 mm (X × Y × Z)

maple-2-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

19. maple-3-separate — cut

Inputs: maple-3

Outputs: maple-3-window, maple-3-reserve

maple-3-window: 108 × 395 × 33 mm (X × Y × Z)

maple-3-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

20. maple-3-end-min — cut

Inputs: maple-3-window

Outputs: maple-3-end1, maple-3-end-min-off

maple-3-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-3-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

21. maple-3-end-max — cut

Inputs: maple-3-end1

Outputs: maple-3-blank, maple-3-end-max-off

maple-3-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-3-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

22. maple-3-zmin — surface

Inputs: maple-3-blank

Outputs: maple-3-z1, maple-3-zmin-loss

maple-3-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-3-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

23. maple-3-zmax — surface

Inputs: maple-3-z1

Outputs: maple-3-z2, maple-3-zmax-loss

maple-3-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-3-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

24. maple-3-xmin — surface

Inputs: maple-3-z2

Outputs: maple-3-x1, maple-3-xmin-loss

maple-3-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-3-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

25. maple-3-xmax — surface

Inputs: maple-3-x1

Outputs: maple-3-x2, maple-3-xmax-loss

maple-3-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-3-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

26. maple-3-rip — cut

Inputs: maple-3-x2

Outputs: maple-3-pre, maple-3-rip-off

maple-3-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-3-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

27. maple-3-joint — surface

Inputs: maple-3-pre

Outputs: panel-1-strip-2, maple-3-joint-loss

panel-1-strip-2: 25 × 370 × 27 mm (X × Y × Z)

maple-3-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

28. maple-4-separate — cut

Inputs: maple-4

Outputs: maple-4-window, maple-4-reserve

maple-4-window: 108 × 395 × 33 mm (X × Y × Z)

maple-4-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

29. maple-4-end-min — cut

Inputs: maple-4-window

Outputs: maple-4-end1, maple-4-end-min-off

maple-4-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-4-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

30. maple-4-end-max — cut

Inputs: maple-4-end1

Outputs: maple-4-blank, maple-4-end-max-off

maple-4-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-4-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

31. maple-4-zmin — surface

Inputs: maple-4-blank

Outputs: maple-4-z1, maple-4-zmin-loss

maple-4-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-4-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

32. maple-4-zmax — surface

Inputs: maple-4-z1

Outputs: maple-4-z2, maple-4-zmax-loss

maple-4-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-4-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

33. maple-4-xmin — surface

Inputs: maple-4-z2

Outputs: maple-4-x1, maple-4-xmin-loss

maple-4-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-4-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

34. maple-4-xmax — surface

Inputs: maple-4-x1

Outputs: maple-4-x2, maple-4-xmax-loss

maple-4-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-4-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

35. maple-4-rip — cut

Inputs: maple-4-x2

Outputs: maple-4-pre, maple-4-rip-off

maple-4-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-4-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

36. maple-4-joint — surface

Inputs: maple-4-pre

Outputs: panel-1-strip-3, maple-4-joint-loss

panel-1-strip-3: 25 × 370 × 27 mm (X × Y × Z)

maple-4-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

37. maple-5-separate — cut

Inputs: maple-5

Outputs: maple-5-window, maple-5-reserve

maple-5-window: 108 × 395 × 33 mm (X × Y × Z)

maple-5-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

38. maple-5-end-min — cut

Inputs: maple-5-window

Outputs: maple-5-end1, maple-5-end-min-off

maple-5-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-5-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

39. maple-5-end-max — cut

Inputs: maple-5-end1

Outputs: maple-5-blank, maple-5-end-max-off

maple-5-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-5-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

40. maple-5-zmin — surface

Inputs: maple-5-blank

Outputs: maple-5-z1, maple-5-zmin-loss

maple-5-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-5-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

41. maple-5-zmax — surface

Inputs: maple-5-z1

Outputs: maple-5-z2, maple-5-zmax-loss

maple-5-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-5-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

42. maple-5-xmin — surface

Inputs: maple-5-z2

Outputs: maple-5-x1, maple-5-xmin-loss

maple-5-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-5-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

43. maple-5-xmax — surface

Inputs: maple-5-x1

Outputs: maple-5-x2, maple-5-xmax-loss

maple-5-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-5-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

44. maple-5-rip — cut

Inputs: maple-5-x2

Outputs: maple-5-pre, maple-5-rip-off

maple-5-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-5-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

45. maple-5-joint — surface

Inputs: maple-5-pre

Outputs: panel-1-strip-4, maple-5-joint-loss

panel-1-strip-4: 25 × 370 × 27 mm (X × Y × Z)

maple-5-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

46. maple-6-separate — cut

Inputs: maple-6

Outputs: maple-6-window, maple-6-reserve

maple-6-window: 108 × 395 × 33 mm (X × Y × Z)

maple-6-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

47. maple-6-end-min — cut

Inputs: maple-6-window

Outputs: maple-6-end1, maple-6-end-min-off

maple-6-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-6-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

48. maple-6-end-max — cut

Inputs: maple-6-end1

Outputs: maple-6-blank, maple-6-end-max-off

maple-6-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-6-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

49. maple-6-zmin — surface

Inputs: maple-6-blank

Outputs: maple-6-z1, maple-6-zmin-loss

maple-6-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-6-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

50. maple-6-zmax — surface

Inputs: maple-6-z1

Outputs: maple-6-z2, maple-6-zmax-loss

maple-6-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-6-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

51. maple-6-xmin — surface

Inputs: maple-6-z2

Outputs: maple-6-x1, maple-6-xmin-loss

maple-6-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-6-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

52. maple-6-xmax — surface

Inputs: maple-6-x1

Outputs: maple-6-x2, maple-6-xmax-loss

maple-6-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-6-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

53. maple-6-rip — cut

Inputs: maple-6-x2

Outputs: maple-6-pre, maple-6-rip-off

maple-6-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-6-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

54. maple-6-joint — surface

Inputs: maple-6-pre

Outputs: panel-1-strip-5, maple-6-joint-loss

panel-1-strip-5: 25 × 370 × 27 mm (X × Y × Z)

maple-6-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

55. maple-7-separate — cut

Inputs: maple-7

Outputs: maple-7-window, maple-7-reserve

maple-7-window: 108 × 395 × 33 mm (X × Y × Z)

maple-7-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

56. maple-7-end-min — cut

Inputs: maple-7-window

Outputs: maple-7-end1, maple-7-end-min-off

maple-7-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-7-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}
```

57. maple-7-end-max — cut

Inputs: maple-7-end1

Outputs: maple-7-blank, maple-7-end-max-off

maple-7-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-7-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}
```

58. maple-7-zmin — surface

Inputs: maple-7-blank

Outputs: maple-7-z1, maple-7-zmin-loss

maple-7-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-7-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}
```

59. maple-7-zmax — surface

Inputs: maple-7-z1

Outputs: maple-7-z2, maple-7-zmax-loss

maple-7-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-7-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}
```

60. maple-7-xmin — surface

Inputs: maple-7-z2

Outputs: maple-7-x1, maple-7-xmin-loss

maple-7-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-7-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}
```

61. maple-7-xmax — surface

Inputs: maple-7-x1

Outputs: maple-7-x2, maple-7-xmax-loss

maple-7-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-7-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}
```

62. maple-7-rip — cut

Inputs: maple-7-x2

Outputs: maple-7-pre, maple-7-rip-off

maple-7-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-7-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

63. maple-7-joint — surface

Inputs: maple-7-pre

Outputs: panel-1-strip-6, maple-7-joint-loss

panel-1-strip-6: 25 × 370 × 27 mm (X × Y × Z)

maple-7-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

64. maple-8-separate — cut

Inputs: maple-8

Outputs: maple-8-window, maple-8-reserve

maple-8-window: 108 × 395 × 33 mm (X × Y × Z)

maple-8-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

65. maple-8-end-min — cut

Inputs: maple-8-window

Outputs: maple-8-end1, maple-8-end-min-off

maple-8-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-8-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

66. maple-8-end-max — cut

Inputs: maple-8-end1

Outputs: maple-8-blank, maple-8-end-max-off

maple-8-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-8-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

67. maple-8-zmin — surface

Inputs: maple-8-blank

Outputs: maple-8-z1, maple-8-zmin-loss

maple-8-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-8-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

68. maple-8-zmax — surface

Inputs: maple-8-z1

Outputs: maple-8-z2, maple-8-zmax-loss

maple-8-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-8-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

69. maple-8-xmin — surface

Inputs: maple-8-z2

Outputs: maple-8-x1, maple-8-xmin-loss

maple-8-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-8-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

70. maple-8-xmax — surface

Inputs: maple-8-x1

Outputs: maple-8-x2, maple-8-xmax-loss

maple-8-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-8-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

71. maple-8-rip — cut

Inputs: maple-8-x2

Outputs: maple-8-pre, maple-8-rip-off

maple-8-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-8-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

72. maple-8-joint — surface

Inputs: maple-8-pre

Outputs: panel-1-strip-7, maple-8-joint-loss

panel-1-strip-7: 25 × 370 × 27 mm (X × Y × Z)

maple-8-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

73. maple-9-separate — cut

Inputs: maple-9

Outputs: maple-9-window, maple-9-reserve

maple-9-window: 108 × 395 × 33 mm (X × Y × Z)

maple-9-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

74. maple-9-end-min — cut

Inputs: maple-9-window

Outputs: maple-9-end1, maple-9-end-min-off

maple-9-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-9-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

75. maple-9-end-max — cut

Inputs: maple-9-end1

Outputs: maple-9-blank, maple-9-end-max-off

maple-9-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-9-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

76. maple-9-zmin — surface

Inputs: maple-9-blank

Outputs: maple-9-z1, maple-9-zmin-loss

maple-9-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-9-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

77. maple-9-zmax — surface

Inputs: maple-9-z1

Outputs: maple-9-z2, maple-9-zmax-loss

maple-9-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-9-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

78. maple-9-xmin — surface

Inputs: maple-9-z2

Outputs: maple-9-x1, maple-9-xmin-loss

maple-9-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-9-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

79. maple-9-xmax — surface

Inputs: maple-9-x1

Outputs: maple-9-x2, maple-9-xmax-loss

maple-9-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-9-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

80. maple-9-rip — cut

Inputs: maple-9-x2

Outputs: maple-9-pre, maple-9-rip-off

maple-9-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-9-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 29970000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

81. maple-9-joint — surface

Inputs: maple-9-pre

Outputs: panel-1-strip-8, maple-9-joint-loss

panel-1-strip-8: 25 × 370 × 27 mm (X × Y × Z)

maple-9-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

82. maple-10-separate — cut

Inputs: maple-10

Outputs: maple-10-window, maple-10-reserve

maple-10-window: 108 × 395 × 33 mm (X × Y × Z)

maple-10-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

83. maple-10-end-min — cut

Inputs: maple-10-window

Outputs: maple-10-end1, maple-10-end-min-off

maple-10-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-10-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

84. maple-10-end-max — cut

Inputs: maple-10-end1

Outputs: maple-10-blank, maple-10-end-max-off

maple-10-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-10-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

85. maple-10-zmin — surface

Inputs: maple-10-blank

Outputs: maple-10-z1, maple-10-zmin-loss

maple-10-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-10-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

86. maple-10-zmax — surface

Inputs: maple-10-z1

Outputs: maple-10-z2, maple-10-zmax-loss

maple-10-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-10-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

87. maple-10-xmin — surface

Inputs: maple-10-z2

Outputs: maple-10-x1, maple-10-xmin-loss

maple-10-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-10-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

88. maple-10-xmax — surface

Inputs: maple-10-x1

Outputs: maple-10-x2, maple-10-xmax-loss

maple-10-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-10-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

89. maple-10-rip — cut

Inputs: maple-10-x2

Outputs: maple-10-pre, maple-10-rip-off

maple-10-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-10-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

90. maple-10-joint — surface

Inputs: maple-10-pre

Outputs: panel-1-strip-9, maple-10-joint-loss

panel-1-strip-9: 25 × 370 × 27 mm (X × Y × Z)

maple-10-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

91. maple-11-separate — cut

Inputs: maple-11

Outputs: maple-11-window, maple-11-reserve

maple-11-window: 108 × 395 × 33 mm (X × Y × Z)

maple-11-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

92. maple-11-end-min — cut

Inputs: maple-11-window

Outputs: maple-11-end1, maple-11-end-min-off

maple-11-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-11-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

93. maple-11-end-max — cut

Inputs: maple-11-end1

Outputs: maple-11-blank, maple-11-end-max-off

maple-11-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-11-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

94. maple-11-zmin — surface

Inputs: maple-11-blank

Outputs: maple-11-z1, maple-11-zmin-loss

maple-11-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-11-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

95. maple-11-zmax — surface

Inputs: maple-11-z1

Outputs: maple-11-z2, maple-11-zmax-loss

maple-11-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-11-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

96. maple-11-xmin — surface

Inputs: maple-11-z2

Outputs: maple-11-x1, maple-11-xmin-loss

maple-11-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-11-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

97. maple-11-xmax — surface

Inputs: maple-11-x1

Outputs: maple-11-x2, maple-11-xmax-loss

maple-11-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-11-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

98. maple-11-rip — cut

Inputs: maple-11-x2

Outputs: maple-11-pre, maple-11-rip-off

maple-11-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-11-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

99. maple-11-joint — surface

Inputs: maple-11-pre

Outputs: panel-1-strip-10, maple-11-joint-loss

panel-1-strip-10: 25 × 370 × 27 mm (X × Y × Z)

maple-11-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

100. maple-12-separate — cut

Inputs: maple-12

Outputs: maple-12-window, maple-12-reserve

maple-12-window: 108 × 395 × 33 mm (X × Y × Z)

maple-12-reserve: 108 × 21 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 395000, "tool": "saw"}

101. maple-12-end-min — cut

Inputs: maple-12-window

Outputs: maple-12-end1, maple-12-end-min-off

maple-12-end1: 108 × 384 × 33 mm (X × Y × Z)

maple-12-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 395000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 384000, "tool": "saw"}

102. maple-12-end-max — cut

Inputs: maple-12-end1

Outputs: maple-12-blank, maple-12-end-max-off

maple-12-blank: 108 × 370 × 33 mm (X × Y × Z)

maple-12-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 384000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 370000, "tool": "saw"}

103. maple-12-zmin — surface

Inputs: maple-12-blank

Outputs: maple-12-z1, maple-12-zmin-loss

maple-12-z1: 108 × 370 × 30 mm (X × Y × Z)

maple-12-zmin-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 33000], "process": "thickness_planer", "side": "min"}

104. maple-12-zmax — surface

Inputs: maple-12-z1

Outputs: maple-12-z2, maple-12-zmax-loss

maple-12-z2: 108 × 370 × 27 mm (X × Y × Z)

maple-12-zmax-loss: 108 × 370 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 370000, 30000], "process": "thickness_planer", "side": "max"}

105. maple-12-xmin — surface

Inputs: maple-12-z2

Outputs: maple-12-x1, maple-12-xmin-loss

maple-12-x1: 107 × 370 × 27 mm (X × Y × Z)

maple-12-xmin-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 370000, 27000], "process": "jointer", "side": "min"}

106. maple-12-xmax — surface

Inputs: maple-12-x1

Outputs: maple-12-x2, maple-12-xmax-loss

maple-12-x2: 106 × 370 × 27 mm (X × Y × Z)

maple-12-xmax-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 370000, 27000], "process": "jointer", "side": "max"}

107. maple-12-rip — cut

Inputs: maple-12-x2

Outputs: maple-12-pre, maple-12-rip-off

maple-12-pre: 26 × 370 × 27 mm (X × Y × Z)

maple-12-rip-off: 77 × 370 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 370000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2997000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

108. maple-12-joint — surface

Inputs: maple-12-pre

Outputs: panel-1-strip-11, maple-12-joint-loss

panel-1-strip-11: 25 × 370 × 27 mm (X × Y × Z)

maple-12-joint-loss: 1 × 370 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 370000, 27000], "process": "jointer", "side": "max"}

109. panel-1-glue — glue

Inputs: panel-1-strip-0, panel-1-strip-1, panel-1-strip-2, panel-1-strip-3, panel-1-strip-4, panel-1-strip-5, panel-1-strip-6, panel-1-strip-7, panel-1-strip-8, panel-1-strip-9, panel-1-strip-10, panel-1-strip-11

Outputs: panel-1-raw

panel-1-raw: 300 × 370 × 27 mm (X × Y × Z)

```
{"axis": 0, "input_sizes_um": {"panel-1-strip-0": [25000, 370000, 27000], "panel-1-strip-1": [25000, 370000, 27000], "panel-1-strip-10": [25000, 370000, 27000], "panel-1-strip-11": [25000, 370000, 27000], "panel-1-strip-2": [25000, 370000, 27000], "panel-1-strip-3": [25000, 370000, 27000], "panel-1-strip-4": [25000, 370000, 27000], "panel-1-strip-5": [25000, 370000, 27000], "panel-1-strip-6": [25000, 370000, 27000], "panel-1-strip-7": [25000, 370000, 27000], "panel-1-strip-8": [25000, 370000, 27000], "panel-1-strip-9": [25000, 370000, 27000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "first"}
```

110. panel-1-zmin — surface

Inputs: panel-1-raw

Outputs: panel-1-z1, panel-1-zmin-loss

panel-1-z1: 300 × 370 × 26 mm (X × Y × Z)

panel-1-zmin-loss: 300 × 370 × 1 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 370000, 27000], "process": "thickness_planer", "side": "min"}
```

111. panel-1-zmax — surface

Inputs: panel-1-z1

Outputs: panel-1-flat, panel-1-zmax-loss

panel-1-flat: 300 × 370 × 25 mm (X × Y × Z)

panel-1-zmax-loss: 300 × 370 × 1 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 370000, 26000], "process": "thickness_planer", "side": "max"}
```

112. panel-1-slice-0 — cut

Inputs: panel-1-flat

Outputs: slice-0, panel-1-remainder-0

slice-0: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-0: 300 × 335 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 370000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

113. row-0-rotate — rotate

Inputs: slice-0

Outputs: row-0

row-0: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

114. panel-1-slice-1 — cut

Inputs: panel-1-remainder-0

Outputs: slice-1, panel-1-remainder-1

slice-1: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-1: 300 × 300 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 335000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

115. row-1-rotate — rotate

Inputs: slice-1

Outputs: row-1

row-1: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

116. panel-1-slice-2 — cut

Inputs: panel-1-remainder-1

Outputs: slice-8, panel-1-remainder-2

slice-8: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-2: 300 × 265 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 300000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

117. row-8-rotate — rotate

Inputs: slice-8

Outputs: row-8

row-8: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

118. panel-1-slice-3 — cut

Inputs: panel-1-remainder-2

Outputs: slice-9, panel-1-remainder-3

slice-9: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-3: 300 × 230 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 265000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

119. row-9-rotate — rotate

Inputs: slice-9

Outputs: row-9

row-9: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

120. panel-1-slice-4 — cut

Inputs: panel-1-remainder-3

Outputs: slice-10, panel-1-remainder-4

slice-10: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-4: 300 × 195 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 230000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

121. row-10-rotate — rotate

Inputs: slice-10

Outputs: row-10

row-10: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

122. panel-1-slice-5 — cut

Inputs: panel-1-remainder-4

Outputs: slice-11, panel-1-remainder-5

slice-11: 300 × 32 × 25 mm (X × Y × Z)

panel-1-remainder-5: 300 × 160 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 195000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

123. row-11-rotate — rotate

Inputs: slice-11

Outputs: row-11

row-11: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

124. maple-13-separate — cut

Inputs: maple-13

Outputs: maple-13-window, maple-13-reserve

maple-13-window: 108 × 290 × 33 mm (X × Y × Z)

maple-13-reserve: 108 × 126 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}
```

125. maple-13-end-min — cut

Inputs: maple-13-window

Outputs: maple-13-end1, maple-13-end-min-off

maple-13-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-13-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}
```

126. maple-13-end-max — cut

Inputs: maple-13-end1

Outputs: maple-13-blank, maple-13-end-max-off

maple-13-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-13-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

127. maple-13-zmin — surface

Inputs: maple-13-blank

Outputs: maple-13-z1, maple-13-zmin-loss

maple-13-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-13-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

128. maple-13-zmax — surface

Inputs: maple-13-z1

Outputs: maple-13-z2, maple-13-zmax-loss

maple-13-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-13-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

129. maple-13-xmin — surface

Inputs: maple-13-z2

Outputs: maple-13-x1, maple-13-xmin-loss

maple-13-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-13-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

130. maple-13-xmax — surface

Inputs: maple-13-x1

Outputs: maple-13-x2, maple-13-xmax-loss

maple-13-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-13-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

131. maple-13-rip — cut

Inputs: maple-13-x2

Outputs: maple-13-pre, maple-13-rip-off

maple-13-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-13-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

132. maple-13-joint — surface

Inputs: maple-13-pre

Outputs: panel-2-strip-0, maple-13-joint-loss

panel-2-strip-0: 25 × 265 × 27 mm (X × Y × Z)

maple-13-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

133. maple-14-separate — cut

Inputs: maple-14

Outputs: maple-14-window, maple-14-reserve

maple-14-window: 108 × 290 × 33 mm (X × Y × Z)

maple-14-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

134. maple-14-end-min — cut

Inputs: maple-14-window

Outputs: maple-14-end1, maple-14-end-min-off

maple-14-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-14-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

135. maple-14-end-max — cut

Inputs: maple-14-end1

Outputs: maple-14-blank, maple-14-end-max-off

maple-14-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-14-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

136. maple-14-zmin — surface

Inputs: maple-14-blank

Outputs: maple-14-z1, maple-14-zmin-loss

maple-14-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-14-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

137. maple-14-zmax — surface

Inputs: maple-14-z1

Outputs: maple-14-z2, maple-14-zmax-loss

maple-14-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-14-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

138. maple-14-xmin — surface

Inputs: maple-14-z2

Outputs: maple-14-x1, maple-14-xmin-loss

maple-14-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-14-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

139. maple-14-xmax — surface

Inputs: maple-14-x1

Outputs: maple-14-x2, maple-14-xmax-loss

maple-14-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-14-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

140. maple-14-rip — cut

Inputs: maple-14-x2

Outputs: maple-14-pre, maple-14-rip-off

maple-14-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-14-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2146500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

141. maple-14-joint — surface

Inputs: maple-14-pre

Outputs: panel-2-strip-1, maple-14-joint-loss

panel-2-strip-1: 25 × 265 × 27 mm (X × Y × Z)

maple-14-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

142. maple-15-separate — cut

Inputs: maple-15

Outputs: maple-15-window, maple-15-reserve

maple-15-window: 108 × 290 × 33 mm (X × Y × Z)

maple-15-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

143. maple-15-end-min — cut

Inputs: maple-15-window

Outputs: maple-15-end1, maple-15-end-min-off

maple-15-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-15-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

144. maple-15-end-max — cut

Inputs: maple-15-end1

Outputs: maple-15-blank, maple-15-end-max-off

maple-15-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-15-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

145. maple-15-zmin — surface

Inputs: maple-15-blank

Outputs: maple-15-z1, maple-15-zmin-loss

maple-15-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-15-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

146. maple-15-zmax — surface

Inputs: maple-15-z1

Outputs: maple-15-z2, maple-15-zmax-loss

maple-15-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-15-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

147. maple-15-xmin — surface

Inputs: maple-15-z2

Outputs: maple-15-x1, maple-15-xmin-loss

maple-15-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-15-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

148. maple-15-xmax — surface

Inputs: maple-15-x1

Outputs: maple-15-x2, maple-15-xmax-loss

maple-15-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-15-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

149. maple-15-rip — cut

Inputs: maple-15-x2

Outputs: maple-15-pre, maple-15-rip-off

maple-15-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-15-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

150. maple-15-joint — surface

Inputs: maple-15-pre

Outputs: panel-2-strip-2, maple-15-joint-loss

panel-2-strip-2: 25 × 265 × 27 mm (X × Y × Z)

maple-15-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

151. maple-16-separate — cut

Inputs: maple-16

Outputs: maple-16-window, maple-16-reserve

maple-16-window: 108 × 290 × 33 mm (X × Y × Z)

maple-16-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

152. maple-16-end-min — cut

Inputs: maple-16-window

Outputs: maple-16-end1, maple-16-end-min-off

maple-16-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-16-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

153. maple-16-end-max — cut

Inputs: maple-16-end1

Outputs: maple-16-blank, maple-16-end-max-off

maple-16-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-16-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

154. maple-16-zmin — surface

Inputs: maple-16-blank

Outputs: maple-16-z1, maple-16-zmin-loss

maple-16-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-16-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

155. maple-16-zmax — surface

Inputs: maple-16-z1

Outputs: maple-16-z2, maple-16-zmax-loss

maple-16-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-16-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

156. maple-16-xmin — surface

Inputs: maple-16-z2

Outputs: maple-16-x1, maple-16-xmin-loss

maple-16-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-16-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

157. maple-16-xmax — surface

Inputs: maple-16-x1

Outputs: maple-16-x2, maple-16-xmax-loss

maple-16-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-16-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

158. maple-16-rip — cut

Inputs: maple-16-x2

Outputs: maple-16-pre, maple-16-rip-off

maple-16-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-16-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2146500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

159. maple-16-joint — surface

Inputs: maple-16-pre

Outputs: panel-2-strip-3, maple-16-joint-loss

panel-2-strip-3: 25 × 265 × 27 mm (X × Y × Z)

maple-16-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

160. walnut-1-separate — cut

Inputs: walnut-1

Outputs: walnut-1-window, walnut-1-reserve

walnut-1-window: 83 × 290 × 37 mm (X × Y × Z)

walnut-1-reserve: 83 × 126 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

161. walnut-1-end-min — cut

Inputs: walnut-1-window

Outputs: walnut-1-end1, walnut-1-end-min-off

walnut-1-end1: 83 × 279 × 37 mm (X × Y × Z)

walnut-1-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 290000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

162. walnut-1-end-max — cut

Inputs: walnut-1-end1

Outputs: walnut-1-blank, walnut-1-end-max-off

walnut-1-blank: 83 × 265 × 37 mm (X × Y × Z)

walnut-1-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 279000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

163. walnut-1-zmin — surface

Inputs: walnut-1-blank

Outputs: walnut-1-z1, walnut-1-zmin-loss

walnut-1-z1: 83 × 265 × 32 mm (X × Y × Z)

walnut-1-zmin-loss: 83 × 265 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 37000], "process": "thickness_planer", "side": "min"}

164. walnut-1-zmax — surface

Inputs: walnut-1-z1

Outputs: walnut-1-z2, walnut-1-zmax-loss

walnut-1-z2: 83 × 265 × 27 mm (X × Y × Z)

walnut-1-zmax-loss: 83 × 265 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 32000], "process": "thickness_planer", "side": "max"}

165. walnut-1-xmin — surface

Inputs: walnut-1-z2

Outputs: walnut-1-x1, walnut-1-xmin-loss

walnut-1-x1: 82 × 265 × 27 mm (X × Y × Z)

walnut-1-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 265000, 27000], "process": "jointer", "side": "min"}

166. walnut-1-xmax — surface

Inputs: walnut-1-x1

Outputs: walnut-1-x2, walnut-1-xmax-loss

walnut-1-x2: 81 × 265 × 27 mm (X × Y × Z)

walnut-1-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 265000, 27000], "process": "jointer", "side": "max"}

167. walnut-1-rip — cut

Inputs: walnut-1-x2

Outputs: walnut-1-pre, walnut-1-rip-off

walnut-1-pre: 26 × 265 × 27 mm (X × Y × Z)

walnut-1-rip-off: 52 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

168. walnut-1-joint — surface

Inputs: walnut-1-pre

Outputs: panel-2-strip-4, walnut-1-joint-loss

panel-2-strip-4: 25 × 265 × 27 mm (X × Y × Z)

walnut-1-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}
```

169. walnut-2-separate — cut

Inputs: walnut-2

Outputs: walnut-2-window, walnut-2-reserve

walnut-2-window: 83 × 290 × 37 mm (X × Y × Z)

walnut-2-reserve: 83 × 126 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}
```

170. walnut-2-end-min — cut

Inputs: walnut-2-window

Outputs: walnut-2-end1, walnut-2-end-min-off

walnut-2-end1: 83 × 279 × 37 mm (X × Y × Z)

walnut-2-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [83000, 290000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}
```

171. walnut-2-end-max — cut

Inputs: walnut-2-end1

Outputs: walnut-2-blank, walnut-2-end-max-off

walnut-2-blank: 83 × 265 × 37 mm (X × Y × Z)

walnut-2-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [83000, 279000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}
```

172. walnut-2-zmin — surface

Inputs: walnut-2-blank

Outputs: walnut-2-z1, walnut-2-zmin-loss

walnut-2-z1: 83 × 265 × 32 mm (X × Y × Z)

walnut-2-zmin-loss: 83 × 265 × 5 mm (X × Y × Z)

```
{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 37000], "process": "thickness_planer", "side": "min"}
```

173. walnut-2-zmax — surface

Inputs: walnut-2-z1

Outputs: walnut-2-z2, walnut-2-zmax-loss

walnut-2-z2: 83 × 265 × 27 mm (X × Y × Z)

walnut-2-zmax-loss: 83 × 265 × 5 mm (X × Y × Z)

```
{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 32000], "process": "thickness_planer", "side": "max"}
```

174. walnut-2-xmin — surface

Inputs: walnut-2-z2

Outputs: walnut-2-x1, walnut-2-xmin-loss

walnut-2-x1: 82 × 265 × 27 mm (X × Y × Z)

walnut-2-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 265000, 27000], "process": "jointer", "side": "min"}

175. walnut-2-xmax — surface

Inputs: walnut-2-x1

Outputs: walnut-2-x2, walnut-2-xmax-loss

walnut-2-x2: 81 × 265 × 27 mm (X × Y × Z)

walnut-2-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 265000, 27000], "process": "jointer", "side": "max"}

176. walnut-2-rip — cut

Inputs: walnut-2-x2

Outputs: walnut-2-pre, walnut-2-rip-off

walnut-2-pre: 26 × 265 × 27 mm (X × Y × Z)

walnut-2-rip-off: 52 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2146500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

177. walnut-2-joint — surface

Inputs: walnut-2-pre

Outputs: panel-2-strip-5, walnut-2-joint-loss

panel-2-strip-5: 25 × 265 × 27 mm (X × Y × Z)

walnut-2-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

178. walnut-3-separate — cut

Inputs: walnut-3

Outputs: walnut-3-window, walnut-3-reserve

walnut-3-window: 83 × 290 × 37 mm (X × Y × Z)

walnut-3-reserve: 83 × 126 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

179. walnut-3-end-min — cut

Inputs: walnut-3-window

Outputs: walnut-3-end1, walnut-3-end-min-off

walnut-3-end1: 83 × 279 × 37 mm (X × Y × Z)

walnut-3-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 290000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

180. walnut-3-end-max — cut

Inputs: walnut-3-end1

Outputs: walnut-3-blank, walnut-3-end-max-off

walnut-3-blank: 83 × 265 × 37 mm (X × Y × Z)

walnut-3-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 279000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

181. walnut-3-zmin — surface

Inputs: walnut-3-blank

Outputs: walnut-3-z1, walnut-3-zmin-loss

walnut-3-z1: 83 × 265 × 32 mm (X × Y × Z)

walnut-3-zmin-loss: 83 × 265 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 37000], "process": "thickness_planer", "side": "min"}

182. walnut-3-zmax — surface

Inputs: walnut-3-z1

Outputs: walnut-3-z2, walnut-3-zmax-loss

walnut-3-z2: 83 × 265 × 27 mm (X × Y × Z)

walnut-3-zmax-loss: 83 × 265 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 32000], "process": "thickness_planer", "side": "max"}

183. walnut-3-xmin — surface

Inputs: walnut-3-z2

Outputs: walnut-3-x1, walnut-3-xmin-loss

walnut-3-x1: 82 × 265 × 27 mm (X × Y × Z)

walnut-3-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 265000, 27000], "process": "jointer", "side": "min"}

184. walnut-3-xmax — surface

Inputs: walnut-3-x1

Outputs: walnut-3-x2, walnut-3-xmax-loss

walnut-3-x2: 81 × 265 × 27 mm (X × Y × Z)

walnut-3-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 265000, 27000], "process": "jointer", "side": "max"}

185. walnut-3-rip — cut

Inputs: walnut-3-x2

Outputs: walnut-3-pre, walnut-3-rip-off

walnut-3-pre: 26 × 265 × 27 mm (X × Y × Z)

walnut-3-rip-off: 52 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

186. walnut-3-joint — surface

Inputs: walnut-3-pre

Outputs: panel-2-strip-6, walnut-3-joint-loss

panel-2-strip-6: 25 × 265 × 27 mm (X × Y × Z)

walnut-3-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}
```

187. walnut-4-separate — cut

Inputs: walnut-4

Outputs: walnut-4-window, walnut-4-reserve

walnut-4-window: 83 × 290 × 37 mm (X × Y × Z)

walnut-4-reserve: 83 × 126 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}
```

188. walnut-4-end-min — cut

Inputs: walnut-4-window

Outputs: walnut-4-end1, walnut-4-end-min-off

walnut-4-end1: 83 × 279 × 37 mm (X × Y × Z)

walnut-4-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [83000, 290000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}
```

189. walnut-4-end-max — cut

Inputs: walnut-4-end1

Outputs: walnut-4-blank, walnut-4-end-max-off

walnut-4-blank: 83 × 265 × 37 mm (X × Y × Z)

walnut-4-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [83000, 279000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}
```

190. walnut-4-zmin — surface

Inputs: walnut-4-blank

Outputs: walnut-4-z1, walnut-4-zmin-loss

walnut-4-z1: 83 × 265 × 32 mm (X × Y × Z)

walnut-4-zmin-loss: 83 × 265 × 5 mm (X × Y × Z)

```
{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 37000], "process": "thickness_planer", "side": "min"}
```

191. walnut-4-zmax — surface

Inputs: walnut-4-z1

Outputs: walnut-4-z2, walnut-4-zmax-loss

walnut-4-z2: 83 × 265 × 27 mm (X × Y × Z)

walnut-4-zmax-loss: 83 × 265 × 5 mm (X × Y × Z)

```
{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 265000, 32000], "process": "thickness_planer", "side": "max"}
```

192. walnut-4-xmin — surface

Inputs: walnut-4-z2

Outputs: walnut-4-x1, walnut-4-xmin-loss

walnut-4-x1: $82 \times 265 \times 27$ mm (X × Y × Z)

walnut-4-xmin-loss: $1 \times 265 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 265000, 27000], "process": "jointer", "side": "min"}

193. walnut-4-xmax — surface

Inputs: walnut-4-x1

Outputs: walnut-4-x2, walnut-4-xmax-loss

walnut-4-x2: $81 \times 265 \times 27$ mm (X × Y × Z)

walnut-4-xmax-loss: $1 \times 265 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 265000, 27000], "process": "jointer", "side": "max"}

194. walnut-4-rip — cut

Inputs: walnut-4-x2

Outputs: walnut-4-pre, walnut-4-rip-off

walnut-4-pre: $26 \times 265 \times 27$ mm (X × Y × Z)

walnut-4-rip-off: $52 \times 265 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2146500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

195. walnut-4-joint — surface

Inputs: walnut-4-pre

Outputs: panel-2-strip-7, walnut-4-joint-loss

panel-2-strip-7: $25 \times 265 \times 27$ mm (X × Y × Z)

walnut-4-joint-loss: $1 \times 265 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

196. maple-17-separate — cut

Inputs: maple-17

Outputs: maple-17-window, maple-17-reserve

maple-17-window: $108 \times 290 \times 33$ mm (X × Y × Z)

maple-17-reserve: $108 \times 126 \times 33$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

197. maple-17-end-min — cut

Inputs: maple-17-window

Outputs: maple-17-end1, maple-17-end-min-off

maple-17-end1: $108 \times 279 \times 33$ mm (X × Y × Z)

maple-17-end-min-off: $108 \times 8 \times 33$ mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

198. maple-17-end-max — cut

Inputs: maple-17-end1

Outputs: maple-17-blank, maple-17-end-max-off

maple-17-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-17-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

199. maple-17-zmin — surface

Inputs: maple-17-blank

Outputs: maple-17-z1, maple-17-zmin-loss

maple-17-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-17-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

200. maple-17-zmax — surface

Inputs: maple-17-z1

Outputs: maple-17-z2, maple-17-zmax-loss

maple-17-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-17-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

201. maple-17-xmin — surface

Inputs: maple-17-z2

Outputs: maple-17-x1, maple-17-xmin-loss

maple-17-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-17-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

202. maple-17-xmax — surface

Inputs: maple-17-x1

Outputs: maple-17-x2, maple-17-xmax-loss

maple-17-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-17-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

203. maple-17-rip — cut

Inputs: maple-17-x2

Outputs: maple-17-pre, maple-17-rip-off

maple-17-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-17-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

204. maple-17-joint — surface

Inputs: maple-17-pre

Outputs: panel-2-strip-8, maple-17-joint-loss

panel-2-strip-8: 25 × 265 × 27 mm (X × Y × Z)

maple-17-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

205. maple-18-separate — cut

Inputs: maple-18

Outputs: maple-18-window, maple-18-reserve

maple-18-window: 108 × 290 × 33 mm (X × Y × Z)

maple-18-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

206. maple-18-end-min — cut

Inputs: maple-18-window

Outputs: maple-18-end1, maple-18-end-min-off

maple-18-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-18-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

207. maple-18-end-max — cut

Inputs: maple-18-end1

Outputs: maple-18-blank, maple-18-end-max-off

maple-18-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-18-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

208. maple-18-zmin — surface

Inputs: maple-18-blank

Outputs: maple-18-z1, maple-18-zmin-loss

maple-18-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-18-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

209. maple-18-zmax — surface

Inputs: maple-18-z1

Outputs: maple-18-z2, maple-18-zmax-loss

maple-18-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-18-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

210. maple-18-xmin — surface

Inputs: maple-18-z2

Outputs: maple-18-x1, maple-18-xmin-loss

maple-18-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-18-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

211. maple-18-xmax — surface

Inputs: maple-18-x1

Outputs: maple-18-x2, maple-18-xmax-loss

maple-18-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-18-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

212. maple-18-rip — cut

Inputs: maple-18-x2

Outputs: maple-18-pre, maple-18-rip-off

maple-18-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-18-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

213. maple-18-joint — surface

Inputs: maple-18-pre

Outputs: panel-2-strip-9, maple-18-joint-loss

panel-2-strip-9: 25 × 265 × 27 mm (X × Y × Z)

maple-18-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

214. maple-19-separate — cut

Inputs: maple-19

Outputs: maple-19-window, maple-19-reserve

maple-19-window: 108 × 290 × 33 mm (X × Y × Z)

maple-19-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

215. maple-19-end-min — cut

Inputs: maple-19-window

Outputs: maple-19-end1, maple-19-end-min-off

maple-19-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-19-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

216. maple-19-end-max — cut

Inputs: maple-19-end1

Outputs: maple-19-blank, maple-19-end-max-off

maple-19-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-19-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

217. maple-19-zmin — surface

Inputs: maple-19-blank

Outputs: maple-19-z1, maple-19-zmin-loss

maple-19-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-19-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

218. maple-19-zmax — surface

Inputs: maple-19-z1

Outputs: maple-19-z2, maple-19-zmax-loss

maple-19-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-19-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

219. maple-19-xmin — surface

Inputs: maple-19-z2

Outputs: maple-19-x1, maple-19-xmin-loss

maple-19-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-19-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

220. maple-19-xmax — surface

Inputs: maple-19-x1

Outputs: maple-19-x2, maple-19-xmax-loss

maple-19-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-19-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

221. maple-19-rip — cut

Inputs: maple-19-x2

Outputs: maple-19-pre, maple-19-rip-off

maple-19-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-19-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 21465000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

222. maple-19-joint — surface

Inputs: maple-19-pre

Outputs: panel-2-strip-10, maple-19-joint-loss

panel-2-strip-10: 25 × 265 × 27 mm (X × Y × Z)

maple-19-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

223. maple-20-separate — cut

Inputs: maple-20

Outputs: maple-20-window, maple-20-reserve

maple-20-window: 108 × 290 × 33 mm (X × Y × Z)

maple-20-reserve: 108 × 126 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}

224. maple-20-end-min — cut

Inputs: maple-20-window

Outputs: maple-20-end1, maple-20-end-min-off

maple-20-end1: 108 × 279 × 33 mm (X × Y × Z)

maple-20-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 290000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 279000, "tool": "saw"}

225. maple-20-end-max — cut

Inputs: maple-20-end1

Outputs: maple-20-blank, maple-20-end-max-off

maple-20-blank: 108 × 265 × 33 mm (X × Y × Z)

maple-20-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 279000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 265000, "tool": "saw"}

226. maple-20-zmin — surface

Inputs: maple-20-blank

Outputs: maple-20-z1, maple-20-zmin-loss

maple-20-z1: 108 × 265 × 30 mm (X × Y × Z)

maple-20-zmin-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 33000], "process": "thickness_planer", "side": "min"}

227. maple-20-zmax — surface

Inputs: maple-20-z1

Outputs: maple-20-z2, maple-20-zmax-loss

maple-20-z2: 108 × 265 × 27 mm (X × Y × Z)

maple-20-zmax-loss: 108 × 265 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 265000, 30000], "process": "thickness_planer", "side": "max"}

228. maple-20-xmin — surface

Inputs: maple-20-z2

Outputs: maple-20-x1, maple-20-xmin-loss

maple-20-x1: 107 × 265 × 27 mm (X × Y × Z)

maple-20-xmin-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 265000, 27000], "process": "jointer", "side": "min"}

229. maple-20-xmax — surface

Inputs: maple-20-x1

Outputs: maple-20-x2, maple-20-xmax-loss

maple-20-x2: 106 × 265 × 27 mm (X × Y × Z)

maple-20-xmax-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 265000, 27000], "process": "jointer", "side": "max"}

230. maple-20-rip — cut

Inputs: maple-20-x2

Outputs: maple-20-pre, maple-20-rip-off

maple-20-pre: 26 × 265 × 27 mm (X × Y × Z)

maple-20-rip-off: 77 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 265000, 27000], "kerf_um": 3000, "kerf_volume_um3": 2146500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

231. maple-20-joint — surface

Inputs: maple-20-pre

Outputs: panel-2-strip-11, maple-20-joint-loss

panel-2-strip-11: 25 × 265 × 27 mm (X × Y × Z)

maple-20-joint-loss: 1 × 265 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 265000, 27000], "process": "jointer", "side": "max"}

232. panel-2-glue — glue

Inputs: panel-2-strip-0, panel-2-strip-1, panel-2-strip-2, panel-2-strip-3, panel-2-strip-4, panel-2-strip-5, panel-2-strip-6, panel-2-strip-7, panel-2-strip-8, panel-2-strip-9, panel-2-strip-10, panel-2-strip-11

Outputs: panel-2-raw

panel-2-raw: 300 × 265 × 27 mm (X × Y × Z)

{"axis": 0, "input_sizes_um": {"panel-2-strip-0": [25000, 265000, 27000], "panel-2-strip-1": [25000, 265000, 27000], "panel-2-strip-10": [25000, 265000, 27000], "panel-2-strip-11": [25000, 265000, 27000], "panel-2-strip-2": [25000, 265000, 27000], "panel-2-strip-3": [25000, 265000, 27000], "panel-2-strip-4": [25000, 265000, 27000], "panel-2-strip-5": [25000, 265000, 27000], "panel-2-strip-6": [25000, 265000, 27000], "panel-2-strip-7": [25000, 265000, 27000], "panel-2-strip-8": [25000, 265000, 27000], "panel-2-strip-9": [25000, 265000, 27000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "first"}

233. panel-2-zmin — surface

Inputs: panel-2-raw

Outputs: panel-2-z1, panel-2-zmin-loss

panel-2-z1: 300 × 265 × 26 mm (X × Y × Z)

panel-2-zmin-loss: 300 × 265 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 265000, 27000], "process": "thickness_planer", "side": "min"}

234. panel-2-zmax — surface

Inputs: panel-2-z1

Outputs: panel-2-flat, panel-2-zmax-loss

panel-2-flat: 300 × 265 × 25 mm (X × Y × Z)

panel-2-zmax-loss: 300 × 265 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 265000, 26000], "process": "thickness_planer", "side": "max"}

235. panel-2-slice-0 — cut

Inputs: panel-2-flat

Outputs: slice-5, panel-2-remainder-0

slice-5: 300 × 32 × 25 mm (X × Y × Z)

panel-2-remainder-0: 300 × 230 × 25 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 265000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}

236. row-5-rotate — rotate

Inputs: slice-5

Outputs: row-5

row-5: 300 × 25 × 32 mm (X × Y × Z)

{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}

237. panel-2-slice-1 — cut

Inputs: panel-2-remainder-0

Outputs: slice-6, panel-2-remainder-1

slice-6: 300 × 32 × 25 mm (X × Y × Z)

panel-2-remainder-1: 300 × 195 × 25 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 230000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}

238. row-6-rotate — rotate

Inputs: slice-6

Outputs: row-6

row-6: 300 × 25 × 32 mm (X × Y × Z)

{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}

239. panel-2-slice-2 — cut

Inputs: panel-2-remainder-1

Outputs: slice-7, panel-2-remainder-2

slice-7: 300 × 32 × 25 mm (X × Y × Z)

panel-2-remainder-2: 300 × 160 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 195000, 25000], "kerf_um": 3000, "kerf_volume_um3": 2250000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

240. row-7-rotate — rotate

Inputs: slice-7

Outputs: row-7

row-7: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

241. maple-21-separate — cut

Inputs: maple-21

Outputs: maple-21-window, maple-21-reserve

maple-21-window: 108 × 220 × 33 mm (X × Y × Z)

maple-21-reserve: 108 × 196 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}
```

242. maple-21-end-min — cut

Inputs: maple-21-window

Outputs: maple-21-end1, maple-21-end-min-off

maple-21-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-21-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}
```

243. maple-21-end-max — cut

Inputs: maple-21-end1

Outputs: maple-21-blank, maple-21-end-max-off

maple-21-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-21-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}
```

244. maple-21-zmin — surface

Inputs: maple-21-blank

Outputs: maple-21-z1, maple-21-zmin-loss

maple-21-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-21-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}
```

245. maple-21-zmax — surface

Inputs: maple-21-z1

Outputs: maple-21-z2, maple-21-zmax-loss

maple-21-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-21-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

246. maple-21-xmin — surface

Inputs: maple-21-z2

Outputs: maple-21-x1, maple-21-xmin-loss

maple-21-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-21-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

247. maple-21-xmax — surface

Inputs: maple-21-x1

Outputs: maple-21-x2, maple-21-xmax-loss

maple-21-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-21-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

248. maple-21-rip — cut

Inputs: maple-21-x2

Outputs: maple-21-pre, maple-21-rip-off

maple-21-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-21-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

249. maple-21-joint — surface

Inputs: maple-21-pre

Outputs: panel-3-strip-0, maple-21-joint-loss

panel-3-strip-0: 25 × 195 × 27 mm (X × Y × Z)

maple-21-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

250. maple-22-separate — cut

Inputs: maple-22

Outputs: maple-22-window, maple-22-reserve

maple-22-window: 108 × 220 × 33 mm (X × Y × Z)

maple-22-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

251. maple-22-end-min — cut

Inputs: maple-22-window

Outputs: maple-22-end1, maple-22-end-min-off

maple-22-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-22-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

252. maple-22-end-max — cut

Inputs: maple-22-end1

Outputs: maple-22-blank, maple-22-end-max-off

maple-22-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-22-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

253. maple-22-zmin — surface

Inputs: maple-22-blank

Outputs: maple-22-z1, maple-22-zmin-loss

maple-22-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-22-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

254. maple-22-zmax — surface

Inputs: maple-22-z1

Outputs: maple-22-z2, maple-22-zmax-loss

maple-22-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-22-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

255. maple-22-xmin — surface

Inputs: maple-22-z2

Outputs: maple-22-x1, maple-22-xmin-loss

maple-22-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-22-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

256. maple-22-xmax — surface

Inputs: maple-22-x1

Outputs: maple-22-x2, maple-22-xmax-loss

maple-22-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-22-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

257. maple-22-rip — cut

Inputs: maple-22-x2

Outputs: maple-22-pre, maple-22-rip-off

maple-22-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-22-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

258. maple-22-joint — surface

Inputs: maple-22-pre

Outputs: panel-3-strip-1, maple-22-joint-loss

panel-3-strip-1: 25 × 195 × 27 mm (X × Y × Z)

maple-22-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

259. maple-23-separate — cut

Inputs: maple-23

Outputs: maple-23-window, maple-23-reserve

maple-23-window: 108 × 220 × 33 mm (X × Y × Z)

maple-23-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

260. maple-23-end-min — cut

Inputs: maple-23-window

Outputs: maple-23-end1, maple-23-end-min-off

maple-23-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-23-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

261. maple-23-end-max — cut

Inputs: maple-23-end1

Outputs: maple-23-blank, maple-23-end-max-off

maple-23-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-23-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

262. maple-23-zmin — surface

Inputs: maple-23-blank

Outputs: maple-23-z1, maple-23-zmin-loss

maple-23-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-23-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

263. maple-23-zmax — surface

Inputs: maple-23-z1

Outputs: maple-23-z2, maple-23-zmax-loss

maple-23-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-23-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

264. maple-23-xmin — surface

Inputs: maple-23-z2

Outputs: maple-23-x1, maple-23-xmin-loss

maple-23-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-23-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

265. maple-23-xmax — surface

Inputs: maple-23-x1

Outputs: maple-23-x2, maple-23-xmax-loss

maple-23-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-23-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

266. maple-23-rip — cut

Inputs: maple-23-x2

Outputs: maple-23-pre, maple-23-rip-off

maple-23-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-23-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

267. maple-23-joint — surface

Inputs: maple-23-pre

Outputs: panel-3-strip-2, maple-23-joint-loss

panel-3-strip-2: 25 × 195 × 27 mm (X × Y × Z)

maple-23-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

268. walnut-5-separate — cut

Inputs: walnut-5

Outputs: walnut-5-window, walnut-5-reserve

walnut-5-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-5-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

269. walnut-5-end-min — cut

Inputs: walnut-5-window

Outputs: walnut-5-end1, walnut-5-end-min-off

walnut-5-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-5-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

270. walnut-5-end-max — cut

Inputs: walnut-5-end1

Outputs: walnut-5-blank, walnut-5-end-max-off

walnut-5-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-5-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

271. walnut-5-zmin — surface

Inputs: walnut-5-blank

Outputs: walnut-5-z1, walnut-5-zmin-loss

walnut-5-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-5-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

272. walnut-5-zmax — surface

Inputs: walnut-5-z1

Outputs: walnut-5-z2, walnut-5-zmax-loss

walnut-5-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-5-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

273. walnut-5-xmin — surface

Inputs: walnut-5-z2

Outputs: walnut-5-x1, walnut-5-xmin-loss

walnut-5-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-5-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

274. walnut-5-xmax — surface

Inputs: walnut-5-x1

Outputs: walnut-5-x2, walnut-5-xmax-loss

walnut-5-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-5-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

275. walnut-5-rip — cut

Inputs: walnut-5-x2

Outputs: walnut-5-pre, walnut-5-rip-off

walnut-5-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-5-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

276. walnut-5-joint — surface

Inputs: walnut-5-pre

Outputs: panel-3-strip-3, walnut-5-joint-loss

panel-3-strip-3: 25 × 195 × 27 mm (X × Y × Z)

walnut-5-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

277. maple-24-separate — cut

Inputs: maple-24

Outputs: maple-24-window, maple-24-reserve

maple-24-window: 108 × 220 × 33 mm (X × Y × Z)

maple-24-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

278. maple-24-end-min — cut

Inputs: maple-24-window

Outputs: maple-24-end1, maple-24-end-min-off

maple-24-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-24-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

279. maple-24-end-max — cut

Inputs: maple-24-end1

Outputs: maple-24-blank, maple-24-end-max-off

maple-24-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-24-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

280. maple-24-zmin — surface

Inputs: maple-24-blank

Outputs: maple-24-z1, maple-24-zmin-loss

maple-24-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-24-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

281. maple-24-zmax — surface

Inputs: maple-24-z1

Outputs: maple-24-z2, maple-24-zmax-loss

maple-24-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-24-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

282. maple-24-xmin — surface

Inputs: maple-24-z2

Outputs: maple-24-x1, maple-24-xmin-loss

maple-24-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-24-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

283. maple-24-xmax — surface

Inputs: maple-24-x1

Outputs: maple-24-x2, maple-24-xmax-loss

maple-24-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-24-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

284. maple-24-rip — cut

Inputs: maple-24-x2

Outputs: maple-24-pre, maple-24-rip-off

maple-24-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-24-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

285. maple-24-joint — surface

Inputs: maple-24-pre

Outputs: panel-3-strip-4, maple-24-joint-loss

panel-3-strip-4: 25 × 195 × 27 mm (X × Y × Z)

maple-24-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

286. maple-25-separate — cut

Inputs: maple-25

Outputs: maple-25-window, maple-25-reserve

maple-25-window: 108 × 220 × 33 mm (X × Y × Z)

maple-25-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

287. maple-25-end-min — cut

Inputs: maple-25-window

Outputs: maple-25-end1, maple-25-end-min-off

maple-25-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-25-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

288. maple-25-end-max — cut

Inputs: maple-25-end1

Outputs: maple-25-blank, maple-25-end-max-off

maple-25-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-25-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

289. maple-25-zmin — surface

Inputs: maple-25-blank

Outputs: maple-25-z1, maple-25-zmin-loss

maple-25-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-25-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

290. maple-25-zmax — surface

Inputs: maple-25-z1

Outputs: maple-25-z2, maple-25-zmax-loss

maple-25-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-25-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

291. maple-25-xmin — surface

Inputs: maple-25-z2

Outputs: maple-25-x1, maple-25-xmin-loss

maple-25-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-25-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

292. maple-25-xmax — surface

Inputs: maple-25-x1

Outputs: maple-25-x2, maple-25-xmax-loss

maple-25-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-25-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

293. maple-25-rip — cut

Inputs: maple-25-x2

Outputs: maple-25-pre, maple-25-rip-off

maple-25-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-25-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

294. maple-25-joint — surface

Inputs: maple-25-pre

Outputs: panel-3-strip-5, maple-25-joint-loss

panel-3-strip-5: 25 × 195 × 27 mm (X × Y × Z)

maple-25-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

295. maple-26-separate — cut

Inputs: maple-26

Outputs: maple-26-window, maple-26-reserve

maple-26-window: 108 × 220 × 33 mm (X × Y × Z)

maple-26-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

296. maple-26-end-min — cut

Inputs: maple-26-window

Outputs: maple-26-end1, maple-26-end-min-off

maple-26-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-26-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

297. maple-26-end-max — cut

Inputs: maple-26-end1

Outputs: maple-26-blank, maple-26-end-max-off

maple-26-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-26-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

298. maple-26-zmin — surface

Inputs: maple-26-blank

Outputs: maple-26-z1, maple-26-zmin-loss

maple-26-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-26-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

299. maple-26-zmax — surface

Inputs: maple-26-z1

Outputs: maple-26-z2, maple-26-zmax-loss

maple-26-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-26-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

300. maple-26-xmin — surface

Inputs: maple-26-z2

Outputs: maple-26-x1, maple-26-xmin-loss

maple-26-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-26-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

301. maple-26-xmax — surface

Inputs: maple-26-x1

Outputs: maple-26-x2, maple-26-xmax-loss

maple-26-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-26-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

302. maple-26-rip — cut

Inputs: maple-26-x2

Outputs: maple-26-pre, maple-26-rip-off

maple-26-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-26-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

303. maple-26-joint — surface

Inputs: maple-26-pre

Outputs: panel-3-strip-6, maple-26-joint-loss

panel-3-strip-6: 25 × 195 × 27 mm (X × Y × Z)

maple-26-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

304. maple-27-separate — cut

Inputs: maple-27

Outputs: maple-27-window, maple-27-reserve

maple-27-window: 108 × 220 × 33 mm (X × Y × Z)

maple-27-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

305. maple-27-end-min — cut

Inputs: maple-27-window

Outputs: maple-27-end1, maple-27-end-min-off

maple-27-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-27-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

306. maple-27-end-max — cut

Inputs: maple-27-end1

Outputs: maple-27-blank, maple-27-end-max-off

maple-27-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-27-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

307. maple-27-zmin — surface

Inputs: maple-27-blank

Outputs: maple-27-z1, maple-27-zmin-loss

maple-27-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-27-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

308. maple-27-zmax — surface

Inputs: maple-27-z1

Outputs: maple-27-z2, maple-27-zmax-loss

maple-27-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-27-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

309. maple-27-xmin — surface

Inputs: maple-27-z2

Outputs: maple-27-x1, maple-27-xmin-loss

maple-27-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-27-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

310. maple-27-xmax — surface

Inputs: maple-27-x1

Outputs: maple-27-x2, maple-27-xmax-loss

maple-27-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-27-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

311. maple-27-rip — cut

Inputs: maple-27-x2

Outputs: maple-27-pre, maple-27-rip-off

maple-27-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-27-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

312. maple-27-joint — surface

Inputs: maple-27-pre

Outputs: panel-3-strip-7, maple-27-joint-loss

panel-3-strip-7: 25 × 195 × 27 mm (X × Y × Z)

maple-27-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

313. walnut-6-separate — cut

Inputs: walnut-6

Outputs: walnut-6-window, walnut-6-reserve

walnut-6-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-6-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

314. walnut-6-end-min — cut

Inputs: walnut-6-window

Outputs: walnut-6-end1, walnut-6-end-min-off

walnut-6-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-6-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

315. walnut-6-end-max — cut

Inputs: walnut-6-end1

Outputs: walnut-6-blank, walnut-6-end-max-off

walnut-6-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-6-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

316. walnut-6-zmin — surface

Inputs: walnut-6-blank

Outputs: walnut-6-z1, walnut-6-zmin-loss

walnut-6-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-6-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

317. walnut-6-zmax — surface

Inputs: walnut-6-z1

Outputs: walnut-6-z2, walnut-6-zmax-loss

walnut-6-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-6-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

318. walnut-6-xmin — surface

Inputs: walnut-6-z2

Outputs: walnut-6-x1, walnut-6-xmin-loss

walnut-6-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-6-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

319. walnut-6-xmax — surface

Inputs: walnut-6-x1

Outputs: walnut-6-x2, walnut-6-xmax-loss

walnut-6-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-6-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

320. walnut-6-rip — cut

Inputs: walnut-6-x2

Outputs: walnut-6-pre, walnut-6-rip-off

walnut-6-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-6-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

321. walnut-6-joint — surface

Inputs: walnut-6-pre

Outputs: panel-3-strip-8, walnut-6-joint-loss

panel-3-strip-8: 25 × 195 × 27 mm (X × Y × Z)

walnut-6-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

322. maple-28-separate — cut

Inputs: maple-28

Outputs: maple-28-window, maple-28-reserve

maple-28-window: 108 × 220 × 33 mm (X × Y × Z)

maple-28-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

323. maple-28-end-min — cut

Inputs: maple-28-window

Outputs: maple-28-end1, maple-28-end-min-off

maple-28-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-28-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}
```

324. maple-28-end-max — cut

Inputs: maple-28-end1

Outputs: maple-28-blank, maple-28-end-max-off

maple-28-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-28-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}
```

325. maple-28-zmin — surface

Inputs: maple-28-blank

Outputs: maple-28-z1, maple-28-zmin-loss

maple-28-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-28-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}
```

326. maple-28-zmax — surface

Inputs: maple-28-z1

Outputs: maple-28-z2, maple-28-zmax-loss

maple-28-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-28-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}
```

327. maple-28-xmin — surface

Inputs: maple-28-z2

Outputs: maple-28-x1, maple-28-xmin-loss

maple-28-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-28-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}
```

328. maple-28-xmax — surface

Inputs: maple-28-x1

Outputs: maple-28-x2, maple-28-xmax-loss

maple-28-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-28-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}
```

329. maple-28-rip — cut

Inputs: maple-28-x2

Outputs: maple-28-pre, maple-28-rip-off

maple-28-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-28-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

330. maple-28-joint — surface

Inputs: maple-28-pre

Outputs: panel-3-strip-9, maple-28-joint-loss

panel-3-strip-9: 25 × 195 × 27 mm (X × Y × Z)

maple-28-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

331. maple-29-separate — cut

Inputs: maple-29

Outputs: maple-29-window, maple-29-reserve

maple-29-window: 108 × 220 × 33 mm (X × Y × Z)

maple-29-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

332. maple-29-end-min — cut

Inputs: maple-29-window

Outputs: maple-29-end1, maple-29-end-min-off

maple-29-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-29-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

333. maple-29-end-max — cut

Inputs: maple-29-end1

Outputs: maple-29-blank, maple-29-end-max-off

maple-29-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-29-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

334. maple-29-zmin — surface

Inputs: maple-29-blank

Outputs: maple-29-z1, maple-29-zmin-loss

maple-29-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-29-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

335. maple-29-zmax — surface

Inputs: maple-29-z1

Outputs: maple-29-z2, maple-29-zmax-loss

maple-29-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-29-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

336. maple-29-xmin — surface

Inputs: maple-29-z2

Outputs: maple-29-x1, maple-29-xmin-loss

maple-29-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-29-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

337. maple-29-xmax — surface

Inputs: maple-29-x1

Outputs: maple-29-x2, maple-29-xmax-loss

maple-29-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-29-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

338. maple-29-rip — cut

Inputs: maple-29-x2

Outputs: maple-29-pre, maple-29-rip-off

maple-29-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-29-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

339. maple-29-joint — surface

Inputs: maple-29-pre

Outputs: panel-3-strip-10, maple-29-joint-loss

panel-3-strip-10: 25 × 195 × 27 mm (X × Y × Z)

maple-29-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

340. maple-30-separate — cut

Inputs: maple-30

Outputs: maple-30-window, maple-30-reserve

maple-30-window: 108 × 220 × 33 mm (X × Y × Z)

maple-30-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

341. maple-30-end-min — cut

Inputs: maple-30-window

Outputs: maple-30-end1, maple-30-end-min-off

maple-30-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-30-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

342. maple-30-end-max — cut

Inputs: maple-30-end1

Outputs: maple-30-blank, maple-30-end-max-off

maple-30-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-30-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

343. maple-30-zmin — surface

Inputs: maple-30-blank

Outputs: maple-30-z1, maple-30-zmin-loss

maple-30-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-30-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

344. maple-30-zmax — surface

Inputs: maple-30-z1

Outputs: maple-30-z2, maple-30-zmax-loss

maple-30-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-30-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

345. maple-30-xmin — surface

Inputs: maple-30-z2

Outputs: maple-30-x1, maple-30-xmin-loss

maple-30-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-30-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

346. maple-30-xmax — surface

Inputs: maple-30-x1

Outputs: maple-30-x2, maple-30-xmax-loss

maple-30-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-30-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

347. maple-30-rip — cut

Inputs: maple-30-x2

Outputs: maple-30-pre, maple-30-rip-off

maple-30-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-30-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

348. maple-30-joint — surface

Inputs: maple-30-pre

Outputs: panel-3-strip-11, maple-30-joint-loss

panel-3-strip-11: 25 × 195 × 27 mm (X × Y × Z)

maple-30-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

349. panel-3-glue — glue

Inputs: panel-3-strip-0, panel-3-strip-1, panel-3-strip-2, panel-3-strip-3, panel-3-strip-4, panel-3-strip-5, panel-3-strip-6, panel-3-strip-7, panel-3-strip-8, panel-3-strip-9, panel-3-strip-10, panel-3-strip-11

Outputs: panel-3-raw

panel-3-raw: 300 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "input_sizes_um": {"panel-3-strip-0": [25000, 195000, 27000], "panel-3-strip-1": [25000, 195000, 27000], "panel-3-strip-10": [25000, 195000, 27000], "panel-3-strip-11": [25000, 195000, 27000], "panel-3-strip-2": [25000, 195000, 27000], "panel-3-strip-3": [25000, 195000, 27000], "panel-3-strip-4": [25000, 195000, 27000], "panel-3-strip-5": [25000, 195000, 27000], "panel-3-strip-6": [25000, 195000, 27000], "panel-3-strip-7": [25000, 195000, 27000], "panel-3-strip-8": [25000, 195000, 27000], "panel-3-strip-9": [25000, 195000, 27000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "first"}

350. panel-3-zmin — surface

Inputs: panel-3-raw

Outputs: panel-3-z1, panel-3-zmin-loss

panel-3-z1: 300 × 195 × 26 mm (X × Y × Z)

panel-3-zmin-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 27000], "process": "thickness_planer", "side": "min"}

351. panel-3-zmax — surface

Inputs: panel-3-z1

Outputs: panel-3-flat, panel-3-zmax-loss

panel-3-flat: 300 × 195 × 25 mm (X × Y × Z)

panel-3-zmax-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 26000], "process": "thickness_planer", "side": "max"}

352. panel-3-slice-0 — cut

Inputs: panel-3-flat

Outputs: slice-2, panel-3-remainder-0

slice-2: 300 × 32 × 25 mm (X × Y × Z)

panel-3-remainder-0: 300 × 160 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 195000, 25000], "kerf_um": 3000, "kerf_volume_um3": 2250000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

353. row-2-rotate — rotate

Inputs: slice-2

Outputs: row-2

row-2: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

354. maple-31-separate — cut

Inputs: maple-31

Outputs: maple-31-window, maple-31-reserve

maple-31-window: 108 × 220 × 33 mm (X × Y × Z)

maple-31-reserve: 108 × 196 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}
```

355. maple-31-end-min — cut

Inputs: maple-31-window

Outputs: maple-31-end1, maple-31-end-min-off

maple-31-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-31-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}
```

356. maple-31-end-max — cut

Inputs: maple-31-end1

Outputs: maple-31-blank, maple-31-end-max-off

maple-31-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-31-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}
```

357. maple-31-zmin — surface

Inputs: maple-31-blank

Outputs: maple-31-z1, maple-31-zmin-loss

maple-31-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-31-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}
```

358. maple-31-zmax — surface

Inputs: maple-31-z1

Outputs: maple-31-z2, maple-31-zmax-loss

maple-31-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-31-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

359. maple-31-xmin — surface

Inputs: maple-31-z2

Outputs: maple-31-x1, maple-31-xmin-loss

maple-31-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-31-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

360. maple-31-xmax — surface

Inputs: maple-31-x1

Outputs: maple-31-x2, maple-31-xmax-loss

maple-31-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-31-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

361. maple-31-rip — cut

Inputs: maple-31-x2

Outputs: maple-31-pre, maple-31-rip-off

maple-31-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-31-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

362. maple-31-joint — surface

Inputs: maple-31-pre

Outputs: panel-4-strip-0, maple-31-joint-loss

panel-4-strip-0: 25 × 195 × 27 mm (X × Y × Z)

maple-31-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

363. maple-32-separate — cut

Inputs: maple-32

Outputs: maple-32-window, maple-32-reserve

maple-32-window: 108 × 220 × 33 mm (X × Y × Z)

maple-32-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

364. maple-32-end-min — cut

Inputs: maple-32-window

Outputs: maple-32-end1, maple-32-end-min-off

maple-32-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-32-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

365. maple-32-end-max — cut

Inputs: maple-32-end1

Outputs: maple-32-blank, maple-32-end-max-off

maple-32-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-32-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

366. maple-32-zmin — surface

Inputs: maple-32-blank

Outputs: maple-32-z1, maple-32-zmin-loss

maple-32-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-32-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

367. maple-32-zmax — surface

Inputs: maple-32-z1

Outputs: maple-32-z2, maple-32-zmax-loss

maple-32-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-32-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

368. maple-32-xmin — surface

Inputs: maple-32-z2

Outputs: maple-32-x1, maple-32-xmin-loss

maple-32-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-32-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

369. maple-32-xmax — surface

Inputs: maple-32-x1

Outputs: maple-32-x2, maple-32-xmax-loss

maple-32-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-32-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

370. maple-32-rip — cut

Inputs: maple-32-x2

Outputs: maple-32-pre, maple-32-rip-off

maple-32-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-32-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

371. maple-32-joint — surface

Inputs: maple-32-pre

Outputs: panel-4-strip-1, maple-32-joint-loss

panel-4-strip-1: 25 × 195 × 27 mm (X × Y × Z)

maple-32-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

372. maple-33-separate — cut

Inputs: maple-33

Outputs: maple-33-window, maple-33-reserve

maple-33-window: 108 × 220 × 33 mm (X × Y × Z)

maple-33-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

373. maple-33-end-min — cut

Inputs: maple-33-window

Outputs: maple-33-end1, maple-33-end-min-off

maple-33-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-33-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

374. maple-33-end-max — cut

Inputs: maple-33-end1

Outputs: maple-33-blank, maple-33-end-max-off

maple-33-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-33-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

375. maple-33-zmin — surface

Inputs: maple-33-blank

Outputs: maple-33-z1, maple-33-zmin-loss

maple-33-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-33-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

376. maple-33-zmax — surface

Inputs: maple-33-z1

Outputs: maple-33-z2, maple-33-zmax-loss

maple-33-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-33-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

377. maple-33-xmin — surface

Inputs: maple-33-z2

Outputs: maple-33-x1, maple-33-xmin-loss

maple-33-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-33-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

378. maple-33-xmax — surface

Inputs: maple-33-x1

Outputs: maple-33-x2, maple-33-xmax-loss

maple-33-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-33-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

379. maple-33-rip — cut

Inputs: maple-33-x2

Outputs: maple-33-pre, maple-33-rip-off

maple-33-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-33-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

380. maple-33-joint — surface

Inputs: maple-33-pre

Outputs: panel-4-strip-2, maple-33-joint-loss

panel-4-strip-2: 25 × 195 × 27 mm (X × Y × Z)

maple-33-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

381. walnut-7-separate — cut

Inputs: walnut-7

Outputs: walnut-7-window, walnut-7-reserve

walnut-7-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-7-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

382. walnut-7-end-min — cut

Inputs: walnut-7-window

Outputs: walnut-7-end1, walnut-7-end-min-off

walnut-7-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-7-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

383. walnut-7-end-max — cut

Inputs: walnut-7-end1

Outputs: walnut-7-blank, walnut-7-end-max-off

walnut-7-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-7-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

384. walnut-7-zmin — surface

Inputs: walnut-7-blank

Outputs: walnut-7-z1, walnut-7-zmin-loss

walnut-7-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-7-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

385. walnut-7-zmax — surface

Inputs: walnut-7-z1

Outputs: walnut-7-z2, walnut-7-zmax-loss

walnut-7-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-7-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

386. walnut-7-xmin — surface

Inputs: walnut-7-z2

Outputs: walnut-7-x1, walnut-7-xmin-loss

walnut-7-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-7-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

387. walnut-7-xmax — surface

Inputs: walnut-7-x1

Outputs: walnut-7-x2, walnut-7-xmax-loss

walnut-7-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-7-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

388. walnut-7-rip — cut

Inputs: walnut-7-x2

Outputs: walnut-7-pre, walnut-7-rip-off

walnut-7-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-7-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

389. walnut-7-joint — surface

Inputs: walnut-7-pre

Outputs: panel-4-strip-3, walnut-7-joint-loss

panel-4-strip-3: 25 × 195 × 27 mm (X × Y × Z)

walnut-7-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

390. walnut-8-separate — cut

Inputs: walnut-8

Outputs: walnut-8-window, walnut-8-reserve

walnut-8-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-8-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

391. walnut-8-end-min — cut

Inputs: walnut-8-window

Outputs: walnut-8-end1, walnut-8-end-min-off

walnut-8-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-8-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

392. walnut-8-end-max — cut

Inputs: walnut-8-end1

Outputs: walnut-8-blank, walnut-8-end-max-off

walnut-8-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-8-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

393. walnut-8-zmin — surface

Inputs: walnut-8-blank

Outputs: walnut-8-z1, walnut-8-zmin-loss

walnut-8-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-8-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

394. walnut-8-zmax — surface

Inputs: walnut-8-z1

Outputs: walnut-8-z2, walnut-8-zmax-loss

walnut-8-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-8-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

395. walnut-8-xmin — surface

Inputs: walnut-8-z2

Outputs: walnut-8-x1, walnut-8-xmin-loss

walnut-8-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-8-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

396. walnut-8-xmax — surface

Inputs: walnut-8-x1

Outputs: walnut-8-x2, walnut-8-xmax-loss

walnut-8-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-8-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

397. walnut-8-rip — cut

Inputs: walnut-8-x2

Outputs: walnut-8-pre, walnut-8-rip-off

walnut-8-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-8-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

398. walnut-8-joint — surface

Inputs: walnut-8-pre

Outputs: panel-4-strip-4, walnut-8-joint-loss

panel-4-strip-4: 25 × 195 × 27 mm (X × Y × Z)

walnut-8-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

399. walnut-9-separate — cut

Inputs: walnut-9

Outputs: walnut-9-window, walnut-9-reserve

walnut-9-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-9-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

400. walnut-9-end-min — cut

Inputs: walnut-9-window

Outputs: walnut-9-end1, walnut-9-end-min-off

walnut-9-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-9-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

401. walnut-9-end-max — cut

Inputs: walnut-9-end1

Outputs: walnut-9-blank, walnut-9-end-max-off

walnut-9-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-9-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

402. walnut-9-zmin — surface

Inputs: walnut-9-blank

Outputs: walnut-9-z1, walnut-9-zmin-loss

walnut-9-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-9-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

403. walnut-9-zmax — surface

Inputs: walnut-9-z1

Outputs: walnut-9-z2, walnut-9-zmax-loss

walnut-9-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-9-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

404. walnut-9-xmin — surface

Inputs: walnut-9-z2

Outputs: walnut-9-x1, walnut-9-xmin-loss

walnut-9-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-9-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

405. walnut-9-xmax — surface

Inputs: walnut-9-x1

Outputs: walnut-9-x2, walnut-9-xmax-loss

walnut-9-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-9-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

406. walnut-9-rip — cut

Inputs: walnut-9-x2

Outputs: walnut-9-pre, walnut-9-rip-off

walnut-9-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-9-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

407. walnut-9-joint — surface

Inputs: walnut-9-pre

Outputs: panel-4-strip-5, walnut-9-joint-loss

panel-4-strip-5: 25 × 195 × 27 mm (X × Y × Z)

walnut-9-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

408. walnut-10-separate — cut

Inputs: walnut-10

Outputs: walnut-10-window, walnut-10-reserve

walnut-10-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-10-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

409. walnut-10-end-min — cut

Inputs: walnut-10-window

Outputs: walnut-10-end1, walnut-10-end-min-off

walnut-10-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-10-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

410. walnut-10-end-max — cut

Inputs: walnut-10-end1

Outputs: walnut-10-blank, walnut-10-end-max-off

walnut-10-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-10-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

411. walnut-10-zmin — surface

Inputs: walnut-10-blank

Outputs: walnut-10-z1, walnut-10-zmin-loss

walnut-10-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-10-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

412. walnut-10-zmax — surface

Inputs: walnut-10-z1

Outputs: walnut-10-z2, walnut-10-zmax-loss

walnut-10-z2: $83 \times 195 \times 27$ mm (X × Y × Z)

walnut-10-zmax-loss: $83 \times 195 \times 5$ mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

413. walnut-10-xmin — surface

Inputs: walnut-10-z2

Outputs: walnut-10-x1, walnut-10-xmin-loss

walnut-10-x1: $82 \times 195 \times 27$ mm (X × Y × Z)

walnut-10-xmin-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

414. walnut-10-xmax — surface

Inputs: walnut-10-x1

Outputs: walnut-10-x2, walnut-10-xmax-loss

walnut-10-x2: $81 \times 195 \times 27$ mm (X × Y × Z)

walnut-10-xmax-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

415. walnut-10-rip — cut

Inputs: walnut-10-x2

Outputs: walnut-10-pre, walnut-10-rip-off

walnut-10-pre: $26 \times 195 \times 27$ mm (X × Y × Z)

walnut-10-rip-off: $52 \times 195 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 1579500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

416. walnut-10-joint — surface

Inputs: walnut-10-pre

Outputs: panel-4-strip-6, walnut-10-joint-loss

panel-4-strip-6: $25 \times 195 \times 27$ mm (X × Y × Z)

walnut-10-joint-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

417. walnut-11-separate — cut

Inputs: walnut-11

Outputs: walnut-11-window, walnut-11-reserve

walnut-11-window: $83 \times 220 \times 37$ mm (X × Y × Z)

walnut-11-reserve: $83 \times 196 \times 37$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

418. walnut-11-end-min — cut

Inputs: walnut-11-window

Outputs: walnut-11-end1, walnut-11-end-min-off

walnut-11-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-11-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

419. walnut-11-end-max — cut

Inputs: walnut-11-end1

Outputs: walnut-11-blank, walnut-11-end-max-off

walnut-11-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-11-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

420. walnut-11-zmin — surface

Inputs: walnut-11-blank

Outputs: walnut-11-z1, walnut-11-zmin-loss

walnut-11-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-11-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

421. walnut-11-zmax — surface

Inputs: walnut-11-z1

Outputs: walnut-11-z2, walnut-11-zmax-loss

walnut-11-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-11-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

422. walnut-11-xmin — surface

Inputs: walnut-11-z2

Outputs: walnut-11-x1, walnut-11-xmin-loss

walnut-11-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-11-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

423. walnut-11-xmax — surface

Inputs: walnut-11-x1

Outputs: walnut-11-x2, walnut-11-xmax-loss

walnut-11-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-11-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

424. walnut-11-rip — cut

Inputs: walnut-11-x2

Outputs: walnut-11-pre, walnut-11-rip-off

walnut-11-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-11-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

425. walnut-11-joint — surface

Inputs: walnut-11-pre

Outputs: panel-4-strip-7, walnut-11-joint-loss

panel-4-strip-7: 25 × 195 × 27 mm (X × Y × Z)

walnut-11-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

426. walnut-12-separate — cut

Inputs: walnut-12

Outputs: walnut-12-window, walnut-12-reserve

walnut-12-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-12-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 92130000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

427. walnut-12-end-min — cut

Inputs: walnut-12-window

Outputs: walnut-12-end1, walnut-12-end-min-off

walnut-12-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-12-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 92130000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

428. walnut-12-end-max — cut

Inputs: walnut-12-end1

Outputs: walnut-12-blank, walnut-12-end-max-off

walnut-12-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-12-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 92130000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

429. walnut-12-zmin — surface

Inputs: walnut-12-blank

Outputs: walnut-12-z1, walnut-12-zmin-loss

walnut-12-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-12-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

430. walnut-12-zmax — surface

Inputs: walnut-12-z1

Outputs: walnut-12-z2, walnut-12-zmax-loss

walnut-12-z2: $83 \times 195 \times 27$ mm (X × Y × Z)

walnut-12-zmax-loss: $83 \times 195 \times 5$ mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

431. walnut-12-xmin — surface

Inputs: walnut-12-z2

Outputs: walnut-12-x1, walnut-12-xmin-loss

walnut-12-x1: $82 \times 195 \times 27$ mm (X × Y × Z)

walnut-12-xmin-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

432. walnut-12-xmax — surface

Inputs: walnut-12-x1

Outputs: walnut-12-x2, walnut-12-xmax-loss

walnut-12-x2: $81 \times 195 \times 27$ mm (X × Y × Z)

walnut-12-xmax-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

433. walnut-12-rip — cut

Inputs: walnut-12-x2

Outputs: walnut-12-pre, walnut-12-rip-off

walnut-12-pre: $26 \times 195 \times 27$ mm (X × Y × Z)

walnut-12-rip-off: $52 \times 195 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 1579500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

434. walnut-12-joint — surface

Inputs: walnut-12-pre

Outputs: panel-4-strip-8, walnut-12-joint-loss

panel-4-strip-8: $25 \times 195 \times 27$ mm (X × Y × Z)

walnut-12-joint-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

435. maple-34-separate — cut

Inputs: maple-34

Outputs: maple-34-window, maple-34-reserve

maple-34-window: $108 \times 220 \times 33$ mm (X × Y × Z)

maple-34-reserve: $108 \times 196 \times 33$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

436. maple-34-end-min — cut

Inputs: maple-34-window

Outputs: maple-34-end1, maple-34-end-min-off

maple-34-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-34-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

437. maple-34-end-max — cut

Inputs: maple-34-end1

Outputs: maple-34-blank, maple-34-end-max-off

maple-34-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-34-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

438. maple-34-zmin — surface

Inputs: maple-34-blank

Outputs: maple-34-z1, maple-34-zmin-loss

maple-34-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-34-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

439. maple-34-zmax — surface

Inputs: maple-34-z1

Outputs: maple-34-z2, maple-34-zmax-loss

maple-34-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-34-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

440. maple-34-xmin — surface

Inputs: maple-34-z2

Outputs: maple-34-x1, maple-34-xmin-loss

maple-34-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-34-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

441. maple-34-xmax — surface

Inputs: maple-34-x1

Outputs: maple-34-x2, maple-34-xmax-loss

maple-34-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-34-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

442. maple-34-rip — cut

Inputs: maple-34-x2

Outputs: maple-34-pre, maple-34-rip-off

maple-34-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-34-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

443. maple-34-joint — surface

Inputs: maple-34-pre

Outputs: panel-4-strip-9, maple-34-joint-loss

panel-4-strip-9: 25 × 195 × 27 mm (X × Y × Z)

maple-34-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

444. maple-35-separate — cut

Inputs: maple-35

Outputs: maple-35-window, maple-35-reserve

maple-35-window: 108 × 220 × 33 mm (X × Y × Z)

maple-35-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

445. maple-35-end-min — cut

Inputs: maple-35-window

Outputs: maple-35-end1, maple-35-end-min-off

maple-35-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-35-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

446. maple-35-end-max — cut

Inputs: maple-35-end1

Outputs: maple-35-blank, maple-35-end-max-off

maple-35-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-35-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

447. maple-35-zmin — surface

Inputs: maple-35-blank

Outputs: maple-35-z1, maple-35-zmin-loss

maple-35-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-35-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

448. maple-35-zmax — surface

Inputs: maple-35-z1

Outputs: maple-35-z2, maple-35-zmax-loss

maple-35-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-35-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

449. maple-35-xmin — surface

Inputs: maple-35-z2

Outputs: maple-35-x1, maple-35-xmin-loss

maple-35-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-35-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

450. maple-35-xmax — surface

Inputs: maple-35-x1

Outputs: maple-35-x2, maple-35-xmax-loss

maple-35-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-35-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

451. maple-35-rip — cut

Inputs: maple-35-x2

Outputs: maple-35-pre, maple-35-rip-off

maple-35-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-35-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

452. maple-35-joint — surface

Inputs: maple-35-pre

Outputs: panel-4-strip-10, maple-35-joint-loss

panel-4-strip-10: 25 × 195 × 27 mm (X × Y × Z)

maple-35-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

453. maple-36-separate — cut

Inputs: maple-36

Outputs: maple-36-window, maple-36-reserve

maple-36-window: 108 × 220 × 33 mm (X × Y × Z)

maple-36-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

454. maple-36-end-min — cut

Inputs: maple-36-window

Outputs: maple-36-end1, maple-36-end-min-off

maple-36-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-36-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

455. maple-36-end-max — cut

Inputs: maple-36-end1

Outputs: maple-36-blank, maple-36-end-max-off

maple-36-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-36-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

456. maple-36-zmin — surface

Inputs: maple-36-blank

Outputs: maple-36-z1, maple-36-zmin-loss

maple-36-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-36-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

457. maple-36-zmax — surface

Inputs: maple-36-z1

Outputs: maple-36-z2, maple-36-zmax-loss

maple-36-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-36-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

458. maple-36-xmin — surface

Inputs: maple-36-z2

Outputs: maple-36-x1, maple-36-xmin-loss

maple-36-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-36-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

459. maple-36-xmax — surface

Inputs: maple-36-x1

Outputs: maple-36-x2, maple-36-xmax-loss

maple-36-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-36-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

460. maple-36-rip — cut

Inputs: maple-36-x2

Outputs: maple-36-pre, maple-36-rip-off

maple-36-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-36-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

461. maple-36-joint — surface

Inputs: maple-36-pre

Outputs: panel-4-strip-11, maple-36-joint-loss

panel-4-strip-11: 25 × 195 × 27 mm (X × Y × Z)

maple-36-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

462. panel-4-glue — glue

Inputs: panel-4-strip-0, panel-4-strip-1, panel-4-strip-2, panel-4-strip-3, panel-4-strip-4, panel-4-strip-5, panel-4-strip-6, panel-4-strip-7, panel-4-strip-8, panel-4-strip-9, panel-4-strip-10, panel-4-strip-11

Outputs: panel-4-raw

panel-4-raw: 300 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "input_sizes_um": {"panel-4-strip-0": [25000, 195000, 27000], "panel-4-strip-1": [25000, 195000, 27000], "panel-4-strip-10": [25000, 195000, 27000], "panel-4-strip-11": [25000, 195000, 27000], "panel-4-strip-2": [25000, 195000, 27000], "panel-4-strip-3": [25000, 195000, 27000], "panel-4-strip-4": [25000, 195000, 27000], "panel-4-strip-5": [25000, 195000, 27000], "panel-4-strip-6": [25000, 195000, 27000], "panel-4-strip-7": [25000, 195000, 27000], "panel-4-strip-8": [25000, 195000, 27000], "panel-4-strip-9": [25000, 195000, 27000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "first"}

463. panel-4-zmin — surface

Inputs: panel-4-raw

Outputs: panel-4-z1, panel-4-zmin-loss

panel-4-z1: 300 × 195 × 26 mm (X × Y × Z)

panel-4-zmin-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 27000], "process": "thickness_planer", "side": "min"}

464. panel-4-zmax — surface

Inputs: panel-4-z1

Outputs: panel-4-flat, panel-4-zmax-loss

panel-4-flat: 300 × 195 × 25 mm (X × Y × Z)

panel-4-zmax-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 26000], "process": "thickness_planer", "side": "max"}

465. panel-4-slice-0 — cut

Inputs: panel-4-flat

Outputs: slice-4, panel-4-remainder-0

slice-4: 300 × 32 × 25 mm (X × Y × Z)

panel-4-remainder-0: 300 × 160 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 195000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

466. row-4-rotate — rotate

Inputs: slice-4

Outputs: row-4

row-4: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

467. maple-37-separate — cut

Inputs: maple-37

Outputs: maple-37-window, maple-37-reserve

maple-37-window: 108 × 220 × 33 mm (X × Y × Z)

maple-37-reserve: 108 × 196 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}
```

468. maple-37-end-min — cut

Inputs: maple-37-window

Outputs: maple-37-end1, maple-37-end-min-off

maple-37-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-37-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}
```

469. maple-37-end-max — cut

Inputs: maple-37-end1

Outputs: maple-37-blank, maple-37-end-max-off

maple-37-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-37-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}
```

470. maple-37-zmin — surface

Inputs: maple-37-blank

Outputs: maple-37-z1, maple-37-zmin-loss

maple-37-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-37-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}
```

471. maple-37-zmax — surface

Inputs: maple-37-z1

Outputs: maple-37-z2, maple-37-zmax-loss

maple-37-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-37-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

472. maple-37-xmin — surface

Inputs: maple-37-z2

Outputs: maple-37-x1, maple-37-xmin-loss

maple-37-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-37-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

473. maple-37-xmax — surface

Inputs: maple-37-x1

Outputs: maple-37-x2, maple-37-xmax-loss

maple-37-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-37-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

474. maple-37-rip — cut

Inputs: maple-37-x2

Outputs: maple-37-pre, maple-37-rip-off

maple-37-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-37-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

475. maple-37-joint — surface

Inputs: maple-37-pre

Outputs: panel-5-strip-0, maple-37-joint-loss

panel-5-strip-0: 25 × 195 × 27 mm (X × Y × Z)

maple-37-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

476. maple-38-separate — cut

Inputs: maple-38

Outputs: maple-38-window, maple-38-reserve

maple-38-window: 108 × 220 × 33 mm (X × Y × Z)

maple-38-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

477. maple-38-end-min — cut

Inputs: maple-38-window

Outputs: maple-38-end1, maple-38-end-min-off

maple-38-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-38-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

478. maple-38-end-max — cut

Inputs: maple-38-end1

Outputs: maple-38-blank, maple-38-end-max-off

maple-38-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-38-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

479. maple-38-zmin — surface

Inputs: maple-38-blank

Outputs: maple-38-z1, maple-38-zmin-loss

maple-38-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-38-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

480. maple-38-zmax — surface

Inputs: maple-38-z1

Outputs: maple-38-z2, maple-38-zmax-loss

maple-38-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-38-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

481. maple-38-xmin — surface

Inputs: maple-38-z2

Outputs: maple-38-x1, maple-38-xmin-loss

maple-38-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-38-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

482. maple-38-xmax — surface

Inputs: maple-38-x1

Outputs: maple-38-x2, maple-38-xmax-loss

maple-38-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-38-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

483. maple-38-rip — cut

Inputs: maple-38-x2

Outputs: maple-38-pre, maple-38-rip-off

maple-38-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-38-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

484. maple-38-joint — surface

Inputs: maple-38-pre

Outputs: panel-5-strip-1, maple-38-joint-loss

panel-5-strip-1: 25 × 195 × 27 mm (X × Y × Z)

maple-38-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

485. walnut-13-separate — cut

Inputs: walnut-13

Outputs: walnut-13-window, walnut-13-reserve

walnut-13-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-13-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

486. walnut-13-end-min — cut

Inputs: walnut-13-window

Outputs: walnut-13-end1, walnut-13-end-min-off

walnut-13-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-13-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

487. walnut-13-end-max — cut

Inputs: walnut-13-end1

Outputs: walnut-13-blank, walnut-13-end-max-off

walnut-13-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-13-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

488. walnut-13-zmin — surface

Inputs: walnut-13-blank

Outputs: walnut-13-z1, walnut-13-zmin-loss

walnut-13-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-13-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

489. walnut-13-zmax — surface

Inputs: walnut-13-z1

Outputs: walnut-13-z2, walnut-13-zmax-loss

walnut-13-z2: $83 \times 195 \times 27$ mm (X × Y × Z)

walnut-13-zmax-loss: $83 \times 195 \times 5$ mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

490. walnut-13-xmin — surface

Inputs: walnut-13-z2

Outputs: walnut-13-x1, walnut-13-xmin-loss

walnut-13-x1: $82 \times 195 \times 27$ mm (X × Y × Z)

walnut-13-xmin-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

491. walnut-13-xmax — surface

Inputs: walnut-13-x1

Outputs: walnut-13-x2, walnut-13-xmax-loss

walnut-13-x2: $81 \times 195 \times 27$ mm (X × Y × Z)

walnut-13-xmax-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

492. walnut-13-rip — cut

Inputs: walnut-13-x2

Outputs: walnut-13-pre, walnut-13-rip-off

walnut-13-pre: $26 \times 195 \times 27$ mm (X × Y × Z)

walnut-13-rip-off: $52 \times 195 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 1579500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

493. walnut-13-joint — surface

Inputs: walnut-13-pre

Outputs: panel-5-strip-2, walnut-13-joint-loss

panel-5-strip-2: $25 \times 195 \times 27$ mm (X × Y × Z)

walnut-13-joint-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

494. walnut-14-separate — cut

Inputs: walnut-14

Outputs: walnut-14-window, walnut-14-reserve

walnut-14-window: $83 \times 220 \times 37$ mm (X × Y × Z)

walnut-14-reserve: $83 \times 196 \times 37$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

495. walnut-14-end-min — cut

Inputs: walnut-14-window

Outputs: walnut-14-end1, walnut-14-end-min-off

walnut-14-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-14-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

496. walnut-14-end-max — cut

Inputs: walnut-14-end1

Outputs: walnut-14-blank, walnut-14-end-max-off

walnut-14-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-14-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

497. walnut-14-zmin — surface

Inputs: walnut-14-blank

Outputs: walnut-14-z1, walnut-14-zmin-loss

walnut-14-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-14-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

498. walnut-14-zmax — surface

Inputs: walnut-14-z1

Outputs: walnut-14-z2, walnut-14-zmax-loss

walnut-14-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-14-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

499. walnut-14-xmin — surface

Inputs: walnut-14-z2

Outputs: walnut-14-x1, walnut-14-xmin-loss

walnut-14-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-14-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

500. walnut-14-xmax — surface

Inputs: walnut-14-x1

Outputs: walnut-14-x2, walnut-14-xmax-loss

walnut-14-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-14-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

501. walnut-14-rip — cut

Inputs: walnut-14-x2

Outputs: walnut-14-pre, walnut-14-rip-off

walnut-14-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-14-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

502. walnut-14-joint — surface

Inputs: walnut-14-pre

Outputs: panel-5-strip-3, walnut-14-joint-loss

panel-5-strip-3: 25 × 195 × 27 mm (X × Y × Z)

walnut-14-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

503. walnut-15-separate — cut

Inputs: walnut-15

Outputs: walnut-15-window, walnut-15-reserve

walnut-15-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-15-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

504. walnut-15-end-min — cut

Inputs: walnut-15-window

Outputs: walnut-15-end1, walnut-15-end-min-off

walnut-15-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-15-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

505. walnut-15-end-max — cut

Inputs: walnut-15-end1

Outputs: walnut-15-blank, walnut-15-end-max-off

walnut-15-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-15-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

506. walnut-15-zmin — surface

Inputs: walnut-15-blank

Outputs: walnut-15-z1, walnut-15-zmin-loss

walnut-15-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-15-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

507. walnut-15-zmax — surface

Inputs: walnut-15-z1

Outputs: walnut-15-z2, walnut-15-zmax-loss

walnut-15-z2: $83 \times 195 \times 27$ mm (X × Y × Z)

walnut-15-zmax-loss: $83 \times 195 \times 5$ mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

508. walnut-15-xmin — surface

Inputs: walnut-15-z2

Outputs: walnut-15-x1, walnut-15-xmin-loss

walnut-15-x1: $82 \times 195 \times 27$ mm (X × Y × Z)

walnut-15-xmin-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

509. walnut-15-xmax — surface

Inputs: walnut-15-x1

Outputs: walnut-15-x2, walnut-15-xmax-loss

walnut-15-x2: $81 \times 195 \times 27$ mm (X × Y × Z)

walnut-15-xmax-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

510. walnut-15-rip — cut

Inputs: walnut-15-x2

Outputs: walnut-15-pre, walnut-15-rip-off

walnut-15-pre: $26 \times 195 \times 27$ mm (X × Y × Z)

walnut-15-rip-off: $52 \times 195 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 1579500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

511. walnut-15-joint — surface

Inputs: walnut-15-pre

Outputs: panel-5-strip-4, walnut-15-joint-loss

panel-5-strip-4: $25 \times 195 \times 27$ mm (X × Y × Z)

walnut-15-joint-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

512. maple-39-separate — cut

Inputs: maple-39

Outputs: maple-39-window, maple-39-reserve

maple-39-window: $108 \times 220 \times 33$ mm (X × Y × Z)

maple-39-reserve: $108 \times 196 \times 33$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 1069200000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

513. maple-39-end-min — cut

Inputs: maple-39-window

Outputs: maple-39-end1, maple-39-end-min-off

maple-39-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-39-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

514. maple-39-end-max — cut

Inputs: maple-39-end1

Outputs: maple-39-blank, maple-39-end-max-off

maple-39-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-39-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

515. maple-39-zmin — surface

Inputs: maple-39-blank

Outputs: maple-39-z1, maple-39-zmin-loss

maple-39-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-39-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

516. maple-39-zmax — surface

Inputs: maple-39-z1

Outputs: maple-39-z2, maple-39-zmax-loss

maple-39-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-39-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

517. maple-39-xmin — surface

Inputs: maple-39-z2

Outputs: maple-39-x1, maple-39-xmin-loss

maple-39-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-39-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

518. maple-39-xmax — surface

Inputs: maple-39-x1

Outputs: maple-39-x2, maple-39-xmax-loss

maple-39-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-39-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

519. maple-39-rip — cut

Inputs: maple-39-x2

Outputs: maple-39-pre, maple-39-rip-off

maple-39-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-39-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

520. maple-39-joint — surface

Inputs: maple-39-pre

Outputs: panel-5-strip-5, maple-39-joint-loss

panel-5-strip-5: 25 × 195 × 27 mm (X × Y × Z)

maple-39-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

521. maple-40-separate — cut

Inputs: maple-40

Outputs: maple-40-window, maple-40-reserve

maple-40-window: 108 × 220 × 33 mm (X × Y × Z)

maple-40-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

522. maple-40-end-min — cut

Inputs: maple-40-window

Outputs: maple-40-end1, maple-40-end-min-off

maple-40-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-40-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

523. maple-40-end-max — cut

Inputs: maple-40-end1

Outputs: maple-40-blank, maple-40-end-max-off

maple-40-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-40-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

524. maple-40-zmin — surface

Inputs: maple-40-blank

Outputs: maple-40-z1, maple-40-zmin-loss

maple-40-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-40-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

525. maple-40-zmax — surface

Inputs: maple-40-z1

Outputs: maple-40-z2, maple-40-zmax-loss

maple-40-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-40-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

526. maple-40-xmin — surface

Inputs: maple-40-z2

Outputs: maple-40-x1, maple-40-xmin-loss

maple-40-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-40-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

527. maple-40-xmax — surface

Inputs: maple-40-x1

Outputs: maple-40-x2, maple-40-xmax-loss

maple-40-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-40-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

528. maple-40-rip — cut

Inputs: maple-40-x2

Outputs: maple-40-pre, maple-40-rip-off

maple-40-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-40-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

529. maple-40-joint — surface

Inputs: maple-40-pre

Outputs: panel-5-strip-6, maple-40-joint-loss

panel-5-strip-6: 25 × 195 × 27 mm (X × Y × Z)

maple-40-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

530. walnut-16-separate — cut

Inputs: walnut-16

Outputs: walnut-16-window, walnut-16-reserve

walnut-16-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-16-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

531. walnut-16-end-min — cut

Inputs: walnut-16-window

Outputs: walnut-16-end1, walnut-16-end-min-off

walnut-16-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-16-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

532. walnut-16-end-max — cut

Inputs: walnut-16-end1

Outputs: walnut-16-blank, walnut-16-end-max-off

walnut-16-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-16-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

533. walnut-16-zmin — surface

Inputs: walnut-16-blank

Outputs: walnut-16-z1, walnut-16-zmin-loss

walnut-16-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-16-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

534. walnut-16-zmax — surface

Inputs: walnut-16-z1

Outputs: walnut-16-z2, walnut-16-zmax-loss

walnut-16-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-16-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

535. walnut-16-xmin — surface

Inputs: walnut-16-z2

Outputs: walnut-16-x1, walnut-16-xmin-loss

walnut-16-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-16-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

536. walnut-16-xmax — surface

Inputs: walnut-16-x1

Outputs: walnut-16-x2, walnut-16-xmax-loss

walnut-16-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-16-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

537. walnut-16-rip — cut

Inputs: walnut-16-x2

Outputs: walnut-16-pre, walnut-16-rip-off

walnut-16-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-16-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

538. walnut-16-joint — surface

Inputs: walnut-16-pre

Outputs: panel-5-strip-7, walnut-16-joint-loss

panel-5-strip-7: 25 × 195 × 27 mm (X × Y × Z)

walnut-16-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

539. walnut-17-separate — cut

Inputs: walnut-17

Outputs: walnut-17-window, walnut-17-reserve

walnut-17-window: 83 × 220 × 37 mm (X × Y × Z)

walnut-17-reserve: 83 × 196 × 37 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

540. walnut-17-end-min — cut

Inputs: walnut-17-window

Outputs: walnut-17-end1, walnut-17-end-min-off

walnut-17-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-17-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

541. walnut-17-end-max — cut

Inputs: walnut-17-end1

Outputs: walnut-17-blank, walnut-17-end-max-off

walnut-17-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-17-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

542. walnut-17-zmin — surface

Inputs: walnut-17-blank

Outputs: walnut-17-z1, walnut-17-zmin-loss

walnut-17-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-17-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

543. walnut-17-zmax — surface

Inputs: walnut-17-z1

Outputs: walnut-17-z2, walnut-17-zmax-loss

walnut-17-z2: $83 \times 195 \times 27$ mm (X × Y × Z)

walnut-17-zmax-loss: $83 \times 195 \times 5$ mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

544. walnut-17-xmin — surface

Inputs: walnut-17-z2

Outputs: walnut-17-x1, walnut-17-xmin-loss

walnut-17-x1: $82 \times 195 \times 27$ mm (X × Y × Z)

walnut-17-xmin-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

545. walnut-17-xmax — surface

Inputs: walnut-17-x1

Outputs: walnut-17-x2, walnut-17-xmax-loss

walnut-17-x2: $81 \times 195 \times 27$ mm (X × Y × Z)

walnut-17-xmax-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

546. walnut-17-rip — cut

Inputs: walnut-17-x2

Outputs: walnut-17-pre, walnut-17-rip-off

walnut-17-pre: $26 \times 195 \times 27$ mm (X × Y × Z)

walnut-17-rip-off: $52 \times 195 \times 27$ mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 1579500000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

547. walnut-17-joint — surface

Inputs: walnut-17-pre

Outputs: panel-5-strip-8, walnut-17-joint-loss

panel-5-strip-8: $25 \times 195 \times 27$ mm (X × Y × Z)

walnut-17-joint-loss: $1 \times 195 \times 27$ mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

548. walnut-18-separate — cut

Inputs: walnut-18

Outputs: walnut-18-window, walnut-18-reserve

walnut-18-window: $83 \times 220 \times 37$ mm (X × Y × Z)

walnut-18-reserve: $83 \times 196 \times 37$ mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [83000, 419000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

549. walnut-18-end-min — cut

Inputs: walnut-18-window

Outputs: walnut-18-end1, walnut-18-end-min-off

walnut-18-end1: 83 × 209 × 37 mm (X × Y × Z)

walnut-18-end-min-off: 83 × 8 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 220000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

550. walnut-18-end-max — cut

Inputs: walnut-18-end1

Outputs: walnut-18-blank, walnut-18-end-max-off

walnut-18-blank: 83 × 195 × 37 mm (X × Y × Z)

walnut-18-end-max-off: 83 × 11 × 37 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [83000, 209000, 37000], "kerf_um": 3000, "kerf_volume_um3": 9213000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

551. walnut-18-zmin — surface

Inputs: walnut-18-blank

Outputs: walnut-18-z1, walnut-18-zmin-loss

walnut-18-z1: 83 × 195 × 32 mm (X × Y × Z)

walnut-18-zmin-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 37000], "process": "thickness_planer", "side": "min"}

552. walnut-18-zmax — surface

Inputs: walnut-18-z1

Outputs: walnut-18-z2, walnut-18-zmax-loss

walnut-18-z2: 83 × 195 × 27 mm (X × Y × Z)

walnut-18-zmax-loss: 83 × 195 × 5 mm (X × Y × Z)

{"amount_um": 5000, "axis": 2, "category": "milling", "input_size_um": [83000, 195000, 32000], "process": "thickness_planer", "side": "max"}

553. walnut-18-xmin — surface

Inputs: walnut-18-z2

Outputs: walnut-18-x1, walnut-18-xmin-loss

walnut-18-x1: 82 × 195 × 27 mm (X × Y × Z)

walnut-18-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [83000, 195000, 27000], "process": "jointer", "side": "min"}

554. walnut-18-xmax — surface

Inputs: walnut-18-x1

Outputs: walnut-18-x2, walnut-18-xmax-loss

walnut-18-x2: 81 × 195 × 27 mm (X × Y × Z)

walnut-18-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [82000, 195000, 27000], "process": "jointer", "side": "max"}

555. walnut-18-rip — cut

Inputs: walnut-18-x2

Outputs: walnut-18-pre, walnut-18-rip-off

walnut-18-pre: 26 × 195 × 27 mm (X × Y × Z)

walnut-18-rip-off: 52 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [81000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

556. walnut-18-joint — surface

Inputs: walnut-18-pre

Outputs: panel-5-strip-9, walnut-18-joint-loss

panel-5-strip-9: 25 × 195 × 27 mm (X × Y × Z)

walnut-18-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

557. maple-41-separate — cut

Inputs: maple-41

Outputs: maple-41-window, maple-41-reserve

maple-41-window: 108 × 220 × 33 mm (X × Y × Z)

maple-41-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

558. maple-41-end-min — cut

Inputs: maple-41-window

Outputs: maple-41-end1, maple-41-end-min-off

maple-41-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-41-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}

559. maple-41-end-max — cut

Inputs: maple-41-end1

Outputs: maple-41-blank, maple-41-end-max-off

maple-41-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-41-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}

560. maple-41-zmin — surface

Inputs: maple-41-blank

Outputs: maple-41-z1, maple-41-zmin-loss

maple-41-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-41-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}

561. maple-41-zmax — surface

Inputs: maple-41-z1

Outputs: maple-41-z2, maple-41-zmax-loss

maple-41-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-41-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}

562. maple-41-xmin — surface

Inputs: maple-41-z2

Outputs: maple-41-x1, maple-41-xmin-loss

maple-41-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-41-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}

563. maple-41-xmax — surface

Inputs: maple-41-x1

Outputs: maple-41-x2, maple-41-xmax-loss

maple-41-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-41-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}

564. maple-41-rip — cut

Inputs: maple-41-x2

Outputs: maple-41-pre, maple-41-rip-off

maple-41-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-41-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

565. maple-41-joint — surface

Inputs: maple-41-pre

Outputs: panel-5-strip-10, maple-41-joint-loss

panel-5-strip-10: 25 × 195 × 27 mm (X × Y × Z)

maple-41-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

566. maple-42-separate — cut

Inputs: maple-42

Outputs: maple-42-window, maple-42-reserve

maple-42-window: 108 × 220 × 33 mm (X × Y × Z)

maple-42-reserve: 108 × 196 × 33 mm (X × Y × Z)

{"axis": 1, "category": "other_offcuts", "input_size_um": [108000, 419000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 220000, "tool": "saw"}

567. maple-42-end-min — cut

Inputs: maple-42-window

Outputs: maple-42-end1, maple-42-end-min-off

maple-42-end1: 108 × 209 × 33 mm (X × Y × Z)

maple-42-end-min-off: 108 × 8 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 220000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "max", "retained_um": 209000, "tool": "saw"}
```

568. maple-42-end-max — cut

Inputs: maple-42-end1

Outputs: maple-42-blank, maple-42-end-max-off

maple-42-blank: 108 × 195 × 33 mm (X × Y × Z)

maple-42-end-max-off: 108 × 11 × 33 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [108000, 209000, 33000], "kerf_um": 3000, "kerf_volume_um3": 10692000000000, "retained_side": "min", "retained_um": 195000, "tool": "saw"}
```

569. maple-42-zmin — surface

Inputs: maple-42-blank

Outputs: maple-42-z1, maple-42-zmin-loss

maple-42-z1: 108 × 195 × 30 mm (X × Y × Z)

maple-42-zmin-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 33000], "process": "thickness_planer", "side": "min"}
```

570. maple-42-zmax — surface

Inputs: maple-42-z1

Outputs: maple-42-z2, maple-42-zmax-loss

maple-42-z2: 108 × 195 × 27 mm (X × Y × Z)

maple-42-zmax-loss: 108 × 195 × 3 mm (X × Y × Z)

```
{"amount_um": 3000, "axis": 2, "category": "milling", "input_size_um": [108000, 195000, 30000], "process": "thickness_planer", "side": "max"}
```

571. maple-42-xmin — surface

Inputs: maple-42-z2

Outputs: maple-42-x1, maple-42-xmin-loss

maple-42-x1: 107 × 195 × 27 mm (X × Y × Z)

maple-42-xmin-loss: 1 × 195 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [108000, 195000, 27000], "process": "jointer", "side": "min"}
```

572. maple-42-xmax — surface

Inputs: maple-42-x1

Outputs: maple-42-x2, maple-42-xmax-loss

maple-42-x2: 106 × 195 × 27 mm (X × Y × Z)

maple-42-xmax-loss: 1 × 195 × 27 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [107000, 195000, 27000], "process": "jointer", "side": "max"}
```

573. maple-42-rip — cut

Inputs: maple-42-x2

Outputs: maple-42-pre, maple-42-rip-off

maple-42-pre: 26 × 195 × 27 mm (X × Y × Z)

maple-42-rip-off: 77 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "category": "other_offcuts", "input_size_um": [106000, 195000, 27000], "kerf_um": 3000, "kerf_volume_um3": 15795000000000, "retained_side": "min", "retained_um": 26000, "tool": "saw"}

574. maple-42-joint — surface

Inputs: maple-42-pre

Outputs: panel-5-strip-11, maple-42-joint-loss

panel-5-strip-11: 25 × 195 × 27 mm (X × Y × Z)

maple-42-joint-loss: 1 × 195 × 27 mm (X × Y × Z)

{"amount_um": 1000, "axis": 0, "category": "milling", "input_size_um": [26000, 195000, 27000], "process": "jointer", "side": "max"}

575. panel-5-glue — glue

Inputs: panel-5-strip-0, panel-5-strip-1, panel-5-strip-2, panel-5-strip-3, panel-5-strip-4, panel-5-strip-5, panel-5-strip-6, panel-5-strip-7, panel-5-strip-8, panel-5-strip-9, panel-5-strip-10, panel-5-strip-11

Outputs: panel-5-raw

panel-5-raw: 300 × 195 × 27 mm (X × Y × Z)

{"axis": 0, "input_sizes_um": {"panel-5-strip-0": [25000, 195000, 27000], "panel-5-strip-1": [25000, 195000, 27000], "panel-5-strip-10": [25000, 195000, 27000], "panel-5-strip-11": [25000, 195000, 27000], "panel-5-strip-2": [25000, 195000, 27000], "panel-5-strip-3": [25000, 195000, 27000], "panel-5-strip-4": [25000, 195000, 27000], "panel-5-strip-5": [25000, 195000, 27000], "panel-5-strip-6": [25000, 195000, 27000], "panel-5-strip-7": [25000, 195000, 27000], "panel-5-strip-8": [25000, 195000, 27000], "panel-5-strip-9": [25000, 195000, 27000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "first"}

576. panel-5-zmin — surface

Inputs: panel-5-raw

Outputs: panel-5-z1, panel-5-zmin-loss

panel-5-z1: 300 × 195 × 26 mm (X × Y × Z)

panel-5-zmin-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 27000], "process": "thickness_planer", "side": "min"}

577. panel-5-zmax — surface

Inputs: panel-5-z1

Outputs: panel-5-flat, panel-5-zmax-loss

panel-5-flat: 300 × 195 × 25 mm (X × Y × Z)

panel-5-zmax-loss: 300 × 195 × 1 mm (X × Y × Z)

{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 195000, 26000], "process": "thickness_planer", "side": "max"}

578. *panel-5-slice-0 — cut*

Inputs: panel-5-flat

Outputs: slice-3, panel-5-remainder-0

slice-3: 300 × 32 × 25 mm (X × Y × Z)

panel-5-remainder-0: 300 × 160 × 25 mm (X × Y × Z)

```
{"axis": 1, "category": "other_offcuts", "input_size_um": [300000, 195000, 25000], "kerf_um": 3000, "kerf_volume_um3": 22500000000000, "retained_side": "min", "retained_um": 32000, "tool": "saw"}
```

579. *row-3-rotate — rotate*

Inputs: slice-3

Outputs: row-3

row-3: 300 × 25 × 32 mm (X × Y × Z)

```
{"input_size_um": [300000, 32000, 25000], "perm": [0, 2, 1], "sign": [1, -1, 1]}
```

580. *final-glue — glue*

Inputs: row-0, row-1, row-2, row-3, row-4, row-5, row-6, row-7, row-8, row-9, row-10, row-11

Outputs: board-raw

board-raw: 300 × 300 × 32 mm (X × Y × Z)

```
{"axis": 1, "input_sizes_um": {"row-0": [300000, 25000, 32000], "row-1": [300000, 25000, 32000], "row-10": [300000, 25000, 32000], "row-11": [300000, 25000, 32000], "row-2": [300000, 25000, 32000], "row-3": [300000, 25000, 32000], "row-4": [300000, 25000, 32000], "row-5": [300000, 25000, 32000], "row-6": [300000, 25000, 32000], "row-7": [300000, 25000, 32000], "row-8": [300000, 25000, 32000], "row-9": [300000, 25000, 32000]}, "joint_count": 11, "negligible_glue_line": true, "prepared_faces": true, "stage": "final"}
```

581. *final-zmin — surface*

Inputs: board-raw

Outputs: board-z1, final-zmin-loss

board-z1: 300 × 300 × 31 mm (X × Y × Z)

final-zmin-loss: 300 × 300 × 1 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 300000, 32000], "process": "drum_sander", "side": "min"}
```

582. *final-zmax — surface*

Inputs: board-z1

Outputs: board-flat, final-zmax-loss

board-flat: 300 × 300 × 30 mm (X × Y × Z)

final-zmax-loss: 300 × 300 × 1 mm (X × Y × Z)

```
{"amount_um": 1000, "axis": 2, "category": "milling", "input_size_um": [300000, 300000, 31000], "process": "drum_sander", "side": "max"}
```

583. *final-x — cut*

Inputs: board-flat

Outputs: board-x, final-x-off

board-x: 290 × 300 × 30 mm (X × Y × Z)

final-x-off: 7 × 300 × 30 mm (X × Y × Z)

```
{"axis": 0, "category": "trim", "input_size_um": [300000, 300000, 30000], "kerf_um": 3000, "kerf_volume_um3": 27000000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}
```

584. *final-y — cut*

Inputs: board-x

Outputs: finished, final-y-off

finished: 290 × 290 × 30 mm (X × Y × Z)

final-y-off: 290 × 7 × 30 mm (X × Y × Z)

```
{"axis": 1, "category": "trim", "input_size_um": [290000, 300000, 30000], "kerf_um": 3000, "kerf_volume_um3": 26100000000000, "retained_side": "min", "retained_um": 290000, "tool": "saw"}
```